ON-THE-JOB TRAINING SYSTEM FEATURING CENTRALIZED COMPANY HUB WITH STUDENT COMPLETION DATE PREDICTION AND PRACTICUM REPORT GENERATION FOR THE POLYTECHNIC UNIVERSITY OF THE PHILIPPINES – LOPEZ CAMPUS

A Capstone Project

Presented to the Faculty of Information Technology

Polytechnic University of the Philippines

Lopez, Quezon

In Partial Fulfillment of the Requirements

For the Diploma in Information Technology

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TABLE OF CONTENT

LIST OF FIGURES

LIST OF TABLES

CHAPTER 1 – INTRODUCTION

1.1. PROJECT CONTEXT

One of the primary objectives of the University and Colleges is to provide students with a high-quality education and to equip them with the necessary knowledge and skills to excel in their chosen field. This can be achieved through the provision of excellent education, equipment, and quality programs.

At the Polytechnic University of The Philippines (PUP) Lopez Campus, On-The-Job Training (OJT) is one of the academic requirements that students must complete. It provides them with an opportunity to apply classroom knowledge in actual work settings, that allows them to gain hands-on experience and professional exposure. This training also helps students develop professional skills such as communication, teamwork, and time management, which are essential in any workplace. Through this acquisition, students are expected to become better prepared and more equipped to enter the professional world.

However, before the students can begin their OJT, they need to complete the requirements and go through many procedures which include finding their Host Training Establishment (HTE) or the company where they will conduct their practical training.

In line with this, choosing a Host Training Establishment (HTE) requires careful consideration as its nature of business must align with the student's academic program. And based on the result of research questionnaire utilized in this study (see Appendix A), students from Polytechnic University of The Philippines (PUP) Lopez Campus experience difficulty to find a Host Training Establishment (HTE).

In addition to this, each academic program has a required number of hours that need to be rendered during their internship, depending on their curriculum. This requirement plays a crucial role in determining the student's internship completion date. However, many students often miscalculate their expected end date (see Appendix A) due to a lack of proper tracking or confusion to the calculation, which can lead to delays or complications in fulfilling their training requirement.

Moreover, aside from completing the required hours, students are expected to issue a practicum report at the end of their OJT. This report contains the summary of their training experience, the tasks they performed, the skills they acquired during their stay in the company, and other additional requirements set by the university.

From the point of view of Mr. Devimar Mercaida, the OJT Coordinator of PUP Lopez Campus, most of the submitted practicum reports contain several issues. These include grammatical errors, ununiform format, and lack of arrangements of presenting the content. Which all affects the overall quality of the submitted report.

Because of the challenges mentioned, such as finding suitable Host Training Establishment (HTE), miscalculating rendered hours, and the issues in preparing the practicum report, this study was conducted. The main objective of this project is to develop and implement a digital solution that will address these concerns. It aims to provide a system that will assist students throughout their OJT process, from providing an online platform that contains the details of all of the partnered company of Polytechnic University of The Philippines (PUP) Lopez Campus, to a mobile application that could accurately track their rendered hours, predicts their completion date, and helps students to generate and submit well-organized practicum report. The system does not only benefit the students but also the OJT coordinator, as it also provides a dashboard that allows for easier monitoring of student's progress, whether by academic program or by individual.

1.2. TECHNICAL BACKGROUND

1.2.1. Equipment/Hardware

For developers, a personal computer is needed to design, code, test, and debug the system. This computer should be capable of running development tools and frameworks efficiently. The OJT Coordinator may use a personal computer to access the system dashboard, where they can monitor the students' progress and practicum report. While students may use smartphone to download the mobile application for tracking rendered hours, completion date, and generating practicum report.

1.2.2. Software

The system is build using modern and efficient technologies to ensure the quality of the software. Visual Studio Code is used as the text editor to write the code of the system. Postman is also utilized for testing the API services. In terms of

development, in frontend, the developer use React Framework and Tailwind CSS for building and styling the application. Progressive Web App (PWA) is utilized to allows the web application to function as a native
mobile application. On the backend, the server and API is developed using Node.js and Express.js while MongoDB serves as the database where all of the data is manipulated and stored. To incorporate intelligent features for modifying grammars and summarization, the system uses OpenAI. For deployment, the server and API services are hosted in Heroku while the application itself is deployed in Firebase Hosting. Cloudflare is added for security integration and GoDaddy for registering the domain name of the system.

1.2.3. Peopleware/Manpower

To produce quality system requires a good team who could work together with shared goals and commitment. This includes individuals with the skills of frontend development, backend development, UI/UX designing, researching and leadership. Each role contributes to ensures that the system is functional, user-friendly, and aligned with the needs of its intended users.

1.2.4. Network Infrastructure/Architecture

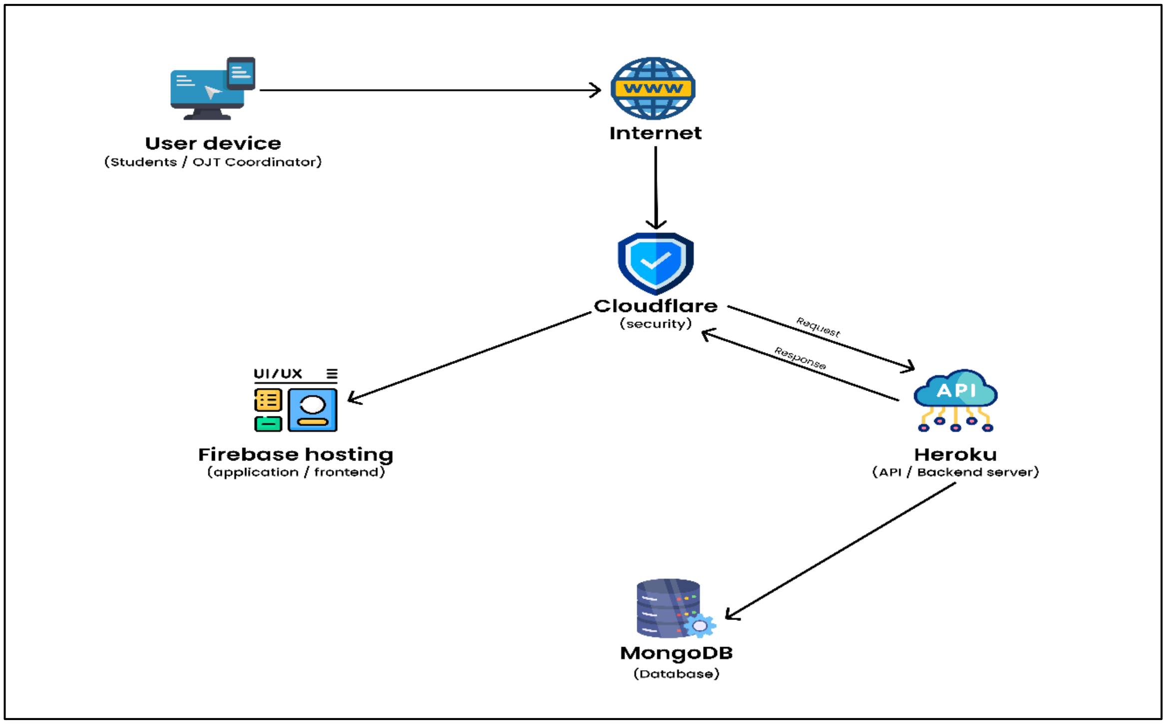

Figure 1. Network Infrastructure/Architecture

The system follows a modern and secure architecture that ensures smooth communication between the end-user and the backend services. Users, such as the students and OJT Coordinator may access the system through their personal devices via the internet. The requests are first routed through Cloudflare that acts as security and performance layer, it provides protection against possible threats such as cyber-attacks. Cloudflare forwards the request either to firebase hosting which serves the frontend of the application, or to Heroku that handles backend server and API services, which processes business logic and handles data interactions by securely communicating with a MongoDB database.

1.2.5. Storage, Backup and Recovery Procedure

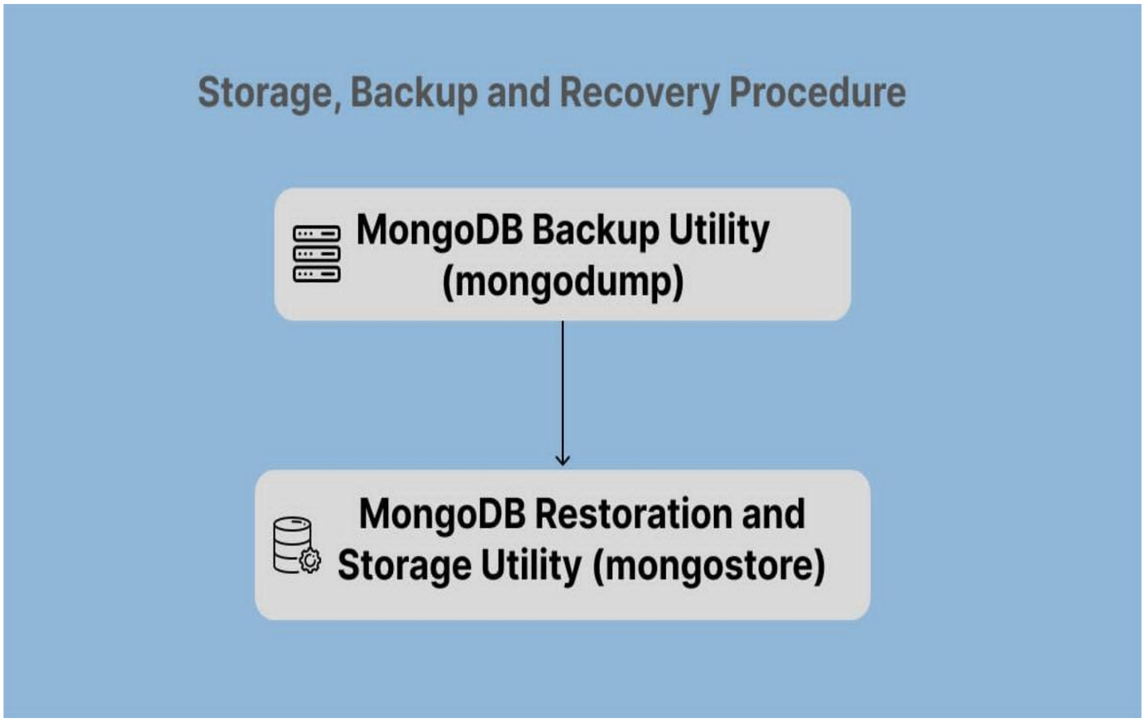

Figure 2. On-The-Job Training system Backup up and Recovery Procedure

The Storage, Backup, and Recovery Procedure for *On-The-Job Training System Featuring a Centralized Company Hub with Student Completion Date Prediction and Practicum Report Generation for the Polytechnic University of the Philippines – Lopez Campus* ensures the integrity of the data and system reliability by using MongoDB Backup Utility (mongodump) and MongoDB Restoration Utility (mongorestore). The backup process involves exporting all database collections into BSON files using **mongodump**, allowing the preservation of essential data such as student records, time logs, and reports. In case of unintentional deletion, corruption, or system failure, **mongorestore** utility enables fast and reliable recovery by importing the backup files back into the system.

1.2.6. Security Procedures

For the system security against possible cyber-attacks, Cloudflare was used as primary security layer. It provides firewall to the system which protects it from malicious activity like DDoS attacks, malicious bots, and other nefarious intrusions. In addition, for external protection, the system implements strong user authentication logic. To be able to access the mobile application, it requires verification from the student's Certificate of Registration (COR), ensuring that only authorized user can log in the system. For administrative access, a dedicated admin account is already provided by the developers to ensure that only authorized personal can manage system settings, monitor student progress, and review practicum reports.

1.2.7. Policies and Procedures

To ensure the secure, efficient, and ethical use of the On-The-Job Training (OJT) System, the following policies and procedures are established:

User Access Policy: Only enrolled students of Polytechnic University of the Philippines – Lopez Campus with a valid Certificate of Registration (COR) are authorized to create an account and access the mobile application. The OJT coordinator is the sole authorized user of the administrator account.

Data Privacy Policy: All student data stored in the system are treated with strict confidentiality. The system complies with the Data Privacy Act, ensuring that no personal or academic information is shared without consent.

System Use Policy: The system must only be used for educational and internship-related purposes. Users are prohibited from using the system for any form of unauthorized access, misuse, or distribution of content.

1.2.8. Data and Process (if applicable)

The system manages various types of data relevant to the On-The-Job Training process. This includes student information (name, student number, academic program, COR), OJT Coordinator profile, and Host Training Establishment (HTE) details (company name, contact details, nature of business), and generated practicum reports. The process begins with the OJT coordinator setting up the OJT Program. After, students may download the mobile application where they can register and verify using their Certificate of Registration (COR). Once registered, students can use the app for time logging, track their progress, and generate their practicum report.

1.3. PROBLEM ANALYSIS

1.3.1. Fishbone Diagram

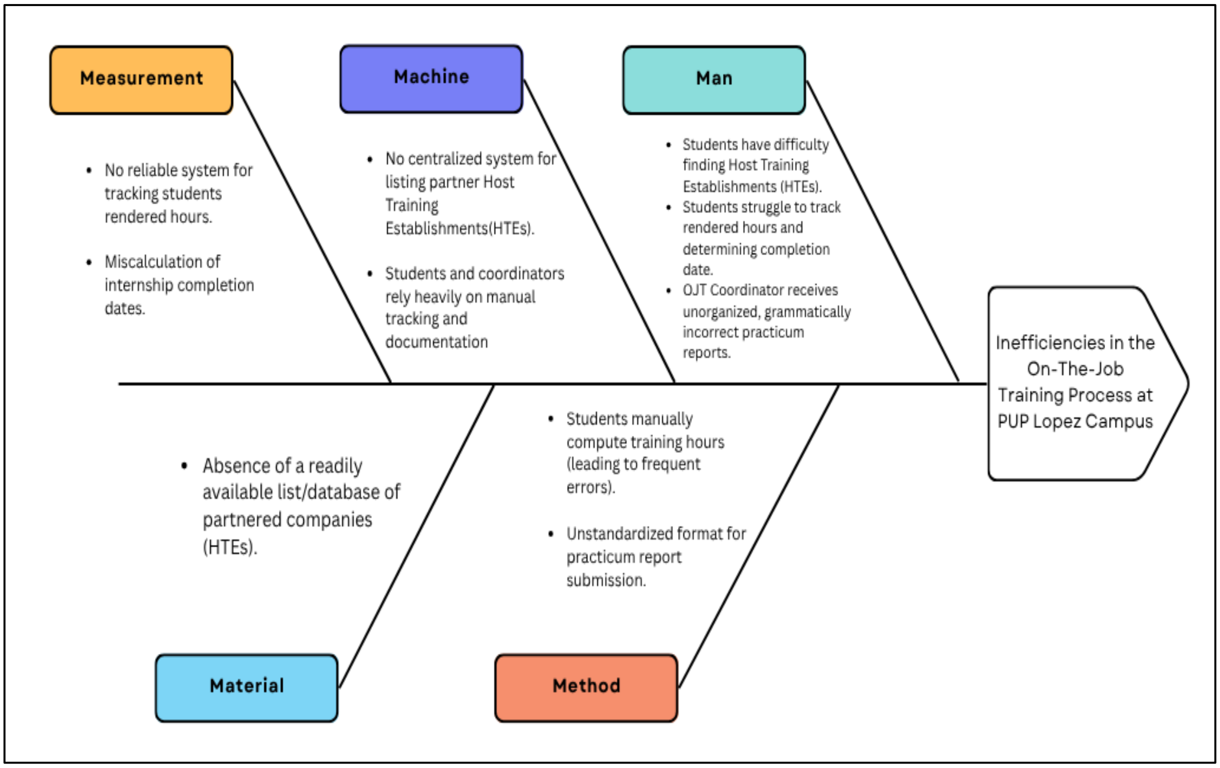

Figure 3. Fishbone Diagram

This Fishbone Diagram illustrates the different reasons caused by the absence of a system in managing the On-The-Job Training (OJT) process at PUP Lopez Campus. It identifies key problem categories such as Man, Machine, Method, Material, and Measurement that contribute to the inefficiencies. Under Man, students face difficulty finding Host Training Establishments (HTEs), struggle with tracking their rendered hours, and submit disorganized practicum reports. Machine issues include the lack of a centralized system and over-reliance on manual processes for tracking and documentation. In terms of Method, students compute training hours manually, often leading to errors, and there is no standard format for submitting practicum reports. The Material category highlights the absence of an accessible list or database of partnered companies. Lastly, under Measurement, there is no reliable system for tracking student hours, which leads to miscalculations in internship completion. These root causes collectively result in delays, confusion, and inefficiencies in the OJT process that can be resolved through a centralized, automated system.

1.3.2. Problem and Solution Statement

Students at Polytechnic University of the Philippines (PUP) Lopez Campus experience difficulty in finding their Host Training Establishment or the company where they will conduct their practical training. Additionally, they struggle with manually tracking their rendered hours and determining their projected completion date, which often leads to delays or complications in fulfilling their training requirement. Lastly, the OJT Coordinator deals with several issues on the submitted practicum report of the students. These include grammatical errors, ununiform format, and lack of arrangements of presenting the content. That is why a digital solution is proposed to address these concerns. The On-The-Job Training system provides online platform that contains the details of all of the partnered company of Polytechnic University of The Philippines (PUP) Lopez Campus, and a mobile application for student that could accurately track their rendered hours, predict their completion date, and assists in generating well-structured practicum reports, streamlining the process for both students and coordinators.

1.3.3. Problem – Requirements Matrix

The Problem–Requirements Matrix outlines how the identified issues in the current On-the-Job Training (OJT) process are addressed by the proposed system through specific features and functionalities.

Problem	Solution
Students face challenges in finding their Host Training Establishment (HTE) where they can complete their required practical training.	Provide centralized company hub that contains the details of all partnered companies of PUP Lopez Campus
Student struggle with manually tracking their rendered hours and determining their projected completion date.	Integrate real-time daily time record which automates the calculation of rendered hours and predicts the completion date.
OJT Coordinator deals with several issues on the submitted practicum report of the students. These include grammatical errors, ununiform format, and lack of arrangements of presenting the content.	Integrate Artificial Intelligent for summarization of the daily reports. And generate uniform and organized practicum report.

Table 1. Problem – Requirements Matrix

1.4. PURPOSE AND DESCRIPTION

The *On-The-Job Training System Featuring a Centralized Company Hub with Student Completion Date Prediction and Practicum Report Generation for the Polytechnic University of the Philippines – Lopez Campus* aims to develop a digital solution to the difficulties and issues experienced by the student and OJT Coordinator at Polytechnic University of The Philippines (PUP) Lopez Campus. The centralized company hub will provide list of partnered companies of PUP Lopez, those that have already been issued a Memorandum of Agreement (MOA) and have previously accommodated OJT students from the university. Students may directly apply to these companies, making the search process easier and more reliable. While the mobile application will help them track their rendered hours and automatically predict the completion date. It is also capable of assisting and generating practicum reports for the students, which also benefits the OJT coordinator as it provide organized format and grammatically correct report.

1.5. SPECIFIC OBJECTIVES

The research aims to fulfill the following objectives:

1. Provide an online accessible partnered company hub of Polytechnic University of The Philippines (PUP) Lopez Campus
2. Develop mobile application that are capable of tracking the rendered time, predict completion date, and generating organized practicum reports of the students.

3. Provide monitoring platform for the OJT coordinator which allow them to easily track and oversee the progress of the students, including its internship status and submitted practicum reports.
4. To design a user-friendly interface that is very easy to use and visually appealing, hence allowing for ease of navigation across different users.

1.6. SCOPE AND LIMITATIONS

This study focuses on the development and implementation of *On-The-Job Training System Featuring a Centralized Company Hub with Student Completion Date Prediction and Practicum Report Generation for the Polytechnic University of the Philippines – Lopez Campus*. It aims to address the problem and issues experienced by the students and OJT coordinator of Polytechnic University of the Philippines – Lopez Campus, which includes a centralized company hub platform for easy access of companies where the students can apply to take their internship. A mobile application for the students capable for tracking their rendered hours and completion date, and also for assisting and generating their practicum reports. Additionally, a monitoring platform is provided to the OJT Coordinator, it is capable for tracking students progress and their submitted practicum reports. However, like any system, it also has some natural limitations and potential areas for future improvement. The centralized company hub only includes official partnered companies of PUP – Lopez Campus so students still need to find alternatives manually if their preferred companies are not on the list. Also, Since the system is web and mobile-based, an active internet connection is required to access features like time logging, company browsing, or generating reports.

CHAPTER 2 – REVIEW OF RELATED LITERATURE/SYSTEMS

This chapter highlights ideas that are significant to the current topic in relation to previous research and is examined to establish the framework for the suggested investigation. The formulation of the study requires a thorough evaluation of the literature and studies to choose the best approach and processes.

On-the-Job Training (OJT) is a concept that allows students to experience actual settings that facilitate the application of theoretical knowledge. It is a necessity incorporated into the curriculum in many institutions including the Polytechnic University of the Philippines – Lopez Campus (PUP-LC) and equips students with the necessary skills for the workplace of his/her chosen discipline. According to Loc et.al (2025), on-the-job training, or OJT, improves employment by promoting industry-specific abilities, boosting adaptability, and encouraging a deeper comprehension of organizational procedures.

In the case of the PUP-LC, regardless of the different programs offered, engineering, business, education, etc., students are required to document their On-the-Job Training experience. This usually means keeping track in documenting their daily activities, attendance, and progress. It becomes, for most of them, a routine that keeps them motivated and ensures they gather the needed training hours for their program. In addition to benefiting the students, as highlighted by Nurfaizi and Hindarto (2023), these tools allow the supervisor to monitor the level of commitment and participation of each intern during the internship experience. In addition, records provide common ground between educational institutions and industries to know the goals of the type of training to be conducted (Technical Education and Skills Development Authority, 2018). The benefit of this format is that PUP can monitor attendance, task completion, and learning outcomes which will all increase employment.

The problem with conventional, manual, on-the-job training, or OJT, tracking like handwritten time sheets and printed reports is that they are inconvenient. Amora et al. (2022) noticed that the manual system of Bohol Island State University is inefficient, subject to error, and susceptible to tampering, resulting in irregularities in records. The OJT AtTRACK System solved these problems through a real time and error-free attendance verification process leveraging IoT and face recognition technology. Likewise, Cabalbag and Ontiveros (2024) pointed out that the paper files of reports in Cagayan State University-Aparri Campus are manual, easily misplaced, and time-consuming because they have to submit hard copy reports. Students sometimes travel very far to submit documents, and the coordinators have an administrative overload managing paperwork, problems that are perhaps replicated in more mass scale in PUP because of its bigger number of students.

Oliveros (2022) pointed out that at Nueva Ecija University of Science and Technology, manually keeping records made the process inconvenient since feedback was delayed and monitoring the progress of the students was difficult. This is further complicated by a lack of consistency in records keeping that affect the reliability of any assessments (Almarinez, 2017). Although Google Drive is currently used at PUP-LC for practicum report submission, the absence of a centralized and automated tracking system may still result in unmet deadlines, misplaced records, or inaccurate hour logs potentially impacting both academic credit and industry evaluations. At a solution, a technology-based solution such as OJT AtTRACK and other similar systems can provide a more efficient, accurate, and real-time monitoring of documentation processes through web-based or IoT platforms (Amora et al., 2022; Cabalbag & Ontiveros, 2024).

Given the increasing complexity of the OJT experience, educational institutions have been moving towards the use of central platforms to help the meeting of students, universities and partnering companies. These systems act as a central interface for the whole range of activities from placing students in internships to monitoring daily actions to assessing companies.

Centralized systems have been pointed out as beneficial to close the gap between academia and industry in several studies. A similar study conducted at Cavite State University – Carmona, a web-based Internship Management System was developed, which served as a centralized hub for document exchanges and communications between all involved parties. Students and the coordinator highly accepted it and it is efficient and accurate in records (Lumbog, 2024).

InternGuide also simplified work processes by automating the relations of students and employers and reducing paperwork; it was developed at the University of Cebu – Lapu-Lapu and Mandaue. Teachers had a simpler time administering it and students found it more intuitive than other systems, according to the feedback received (Aying et al., 2024).

Also, a web-based internship management system has been created to digitize the entire practical training process and enable online access. The iMAPS system has portal facilitates checking internship eligibility, registration, visit scheduling, and oversight of the industrial internship program at Universiti Teknologi MARA (UiTM). This system benefits the educational process for students since they can verify their application status online. In their free time, they are able to concentrate more on their lessons (Jaafar et al., 2017).

This application has several important functional features that provide services to the supervisors and students in the process of OJT for SMK Negeri 1 Boyolali. Among its functionalities is to help field and school supervisors provide better administration of OJT issues like scheduling, student assignment, and overall administration. The system also provides a means of continuous oversight of student training, including attendance and progress towards meeting training obligations. Aside from monitoring, the system also provides extensive reporting, which is important for record-keeping of student progress and performance evaluation (Kurnianingsih, 2022).

Although OJT AtTRACK, InternGuide, and iMAPS have effectively solved problems associated with automation and centralization of on-the-job training documentation and monitoring, the system being proposed for PUP Lopez Campus is more holistic and specific to the context. In addition to the current systems which mainly enhance the tracking of progress and decrease paperwork, this system incorporates three elements that are different: a directory of partnered companies to each academic program that is automatically generated; a calculator of rendered hours with end-date prediction; and a practicum report generator, which aims to standardize report formatting and improve the quality of content. In addition, the system is adapted to the needs and experiences of students and coordinators of the PUP Lopez Campus, providing a mobile accessible and easy to use tool that does more than just monitor but rather helps students pre, during, and post OJT particularly during the placement and documentation process.

CHAPTER 3 – METHODOLOGY

3.1. REQUIREMENTS ANALYSIS

3.1.1. Requirements – Features Matrix

OJT System provides a comprehensive mapping of the functional and non-functional requirements to the corresponding features of the system in this requirements-feature matrix. This ensures that all user needs and system expectations are addressed in the final implementation so as to bring clarity to the minds of the developers and stakeholders. Here is the matrix detailing the requirements and how they are translated into features in the system.

Functional Requirements

Requirements	Features
Coordinator Login	Coordinator can login with authentication to access the allowed part of system.
Coordinator Dashboard and Report	Coordinator panel for monitoring Internship progress, Report per program and section, report per student such as rendered hours and the remaining hours to render and the student's estimated end date.
Coordinator Internship Set Up	Coordinator could set up and update the internship program such as adding which semester the internship will be conducted, and which programs are included.

Coordinator Practicum Report for Monitoring	Coordinator can view and access the practicum report that the student submitted, per program and per student
Student Signup and Login	Students can sign up and login and enter the necessary information.
Student Setup for Internship Details	The student can set up the details by inputting the necessary information such as Company Name, Work Setup, Total hours to render and the Start date of the internship
Student Time Log for Time-in and Time-out	Students panel to record the time-in and time-out, daily report input with real time image capturing.
Student Dashboard for Monitoring	Students can track the rendered hours and remaining hours to render, and see the completion date, the student can also track the percentage of its practicum report progress and can view the real time calendar.
Student Profile	Student can view its personal information and can update the new set of password by clicking the change password button.

Student Practicum Report Generation	Allows the student to input the necessary information and data, attached files to the system, update the practicum report and save the changes,
Centralized Company Hub of PUP Lopez Campus Partnered Companies	Students should be capable of viewing all partnered companies of PUP Lopez Campus and see the details of the internship it offers.
Mobile Compatibility	System optimized for mobile and tablet views to ensure ease of use across devices.

Table 2. Funtional Requirements of Requirements – Features Matrix

Non-Functional Requirements

Requirements	Features
Performance	The system must easily handle high amounts of users as well as uploaded files and images without being affected in its performance.
Scalability	Addition of many students, files, and further features without great architectural changes across the platform
Reliability	The system shall have the capability of minimum amount of downtime such that there exists constant access towards student's file and information.
Security	Incorporate maximum security so all data regarding any student's files and information or other user details such as the coordinator and the company. Implement authentication/encryption.
Cross Platform Compatibility	The system should deliver consistent user experience and functionality across desktops, tables, and smartphones.

Backup and Recovery	It should have regular automated backups and a good ability to recover the system in case of data loss or failure.
Usability	The system should be user-friendly and well-navigated by coordinator and students.

Table 3. Non-Funtional Requirements of Requirements – Features Matrix

3.1.2. Use Case Diagram

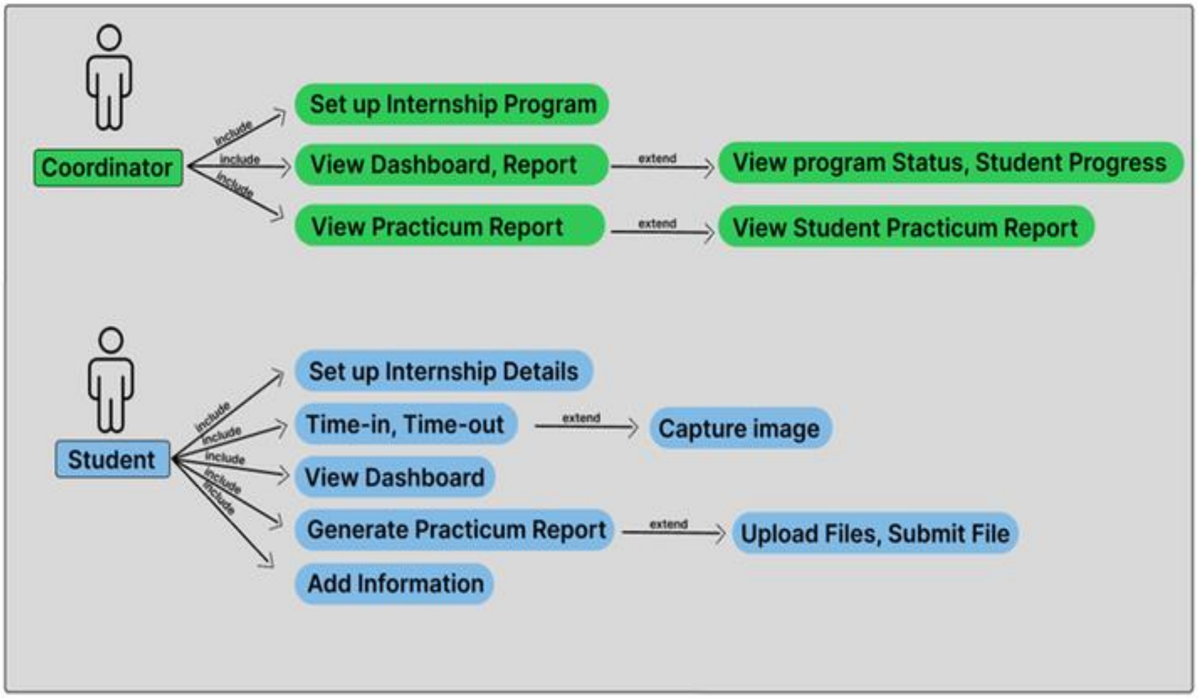

Figure 4. On-The-Job Training System Use Case Diagram

3.1.3. Use Case Report

This case diagram for On-The-Job Training System Featuring a Centralized Company Hub with Student Completion Date Prediction and Practicum Report Generation for the Polytechnic University of the Philippines – Lopez

Campus presents an internship management system with two key user roles: Coordinator and Student. The coordinator can have access to some features such as setting up an internship program which will add the semester and programs that will conduct the internship. The coordinator's “View Dashboard and Report” allows to track the student's progress which includes the student's total rendered hours and remaining hours to render and includes the status of every program's deployment status. The "Setup Internship Program" use case will include the functionality of adding semester and programs that will conduct the internship. The "View Dashboard and Report" is extended by viewing of every program's status of deployment and student's internship progress. Similarly, the "View Practicum Report" use case is extended by "View Student's Practicum Report," which is the submitted accomplished practicum report of a student. Student's "Set up Internship Details" use case includes the functionality of adding details of the student's internship. The “Time-in and Time-out” use case extends the functionality of “Capture Image” which allows to capture the image of the student. The “View Dashboard" includes the internship and practicum progress. The student's “Generate Practicum Report” extends functionality of “Upload files and Submit files” which allows students to upload attached files and submit changes. Both Coordinator and Student can view the internship progress.

3.2. DESIGN SPECIFICATIONS

3.2.1. Activity Diagram (per use case)

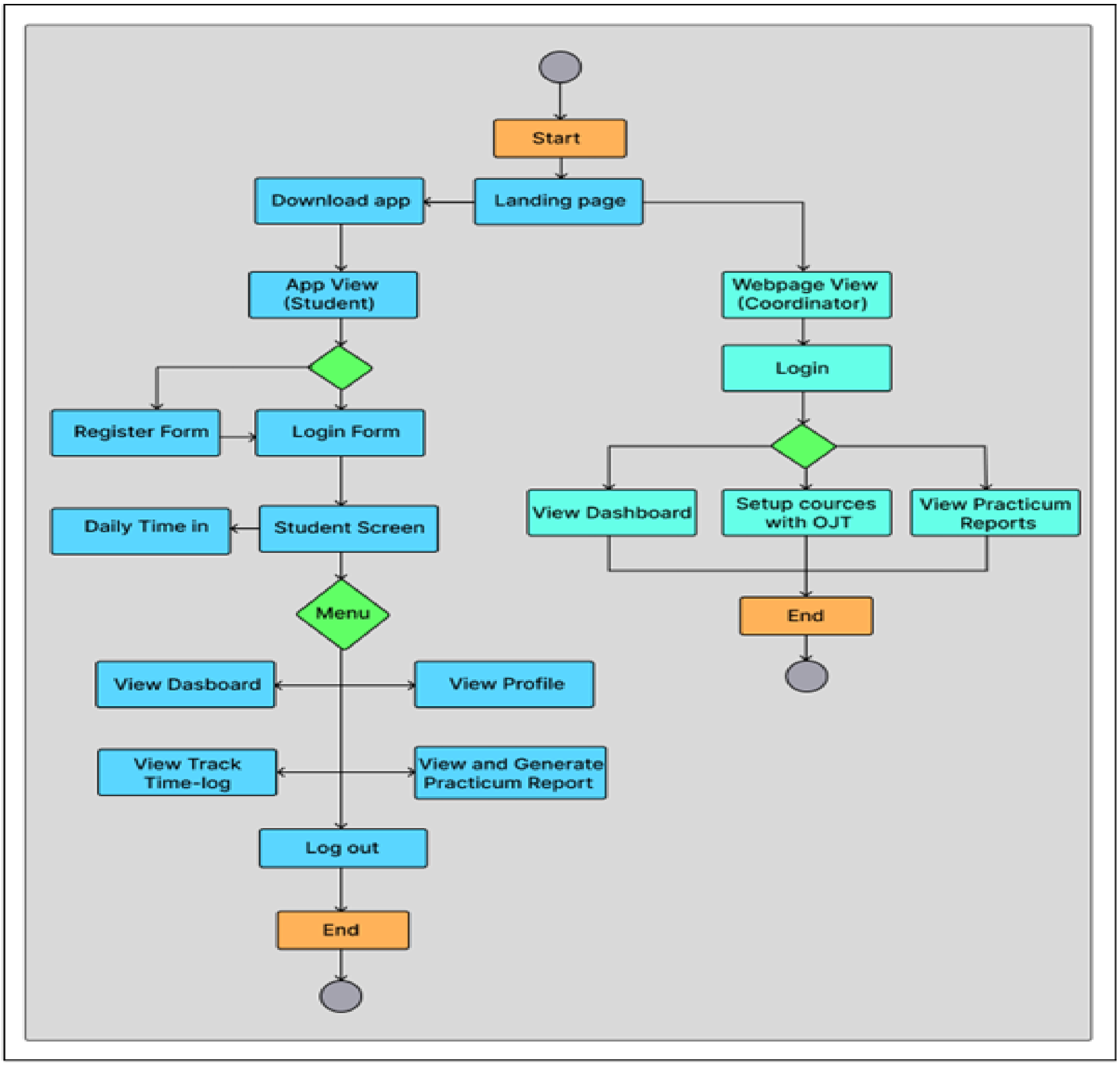

Figure 5. On-The-Job Training System Activity Diagram

The activity diagram of the student and coordinator user experience for an app that is able to handle practicum or on-the-job training (OJT) for students and coordinator. Starting with the process beginning where students and the coordinator first encounter a landing page. From there, student can either download an app to access the system and the coordinator will login through the webpage in the application. The process splits into two streams: students download the app to see the app view of the student, while coordinators access the Webpage View (Coordinator). Students' process is to download the app, followed by a decision to complete a "Register Form" or access the "Login Form" to see the "Student Screen," where they can input "Daily Time In" data and choose a menu option to "View Dashboard," "View Profile," "View Track Time-log," "View and Generate Practicum Report," or "Log out," which ends the session for the student. Coordinators log in via the webpage and can choose to "View Dashboard," "Setup Courses with OJT," or "View Practicum Reports" before reaching the "End" for logging out. art

3.2.2. Class Diagram

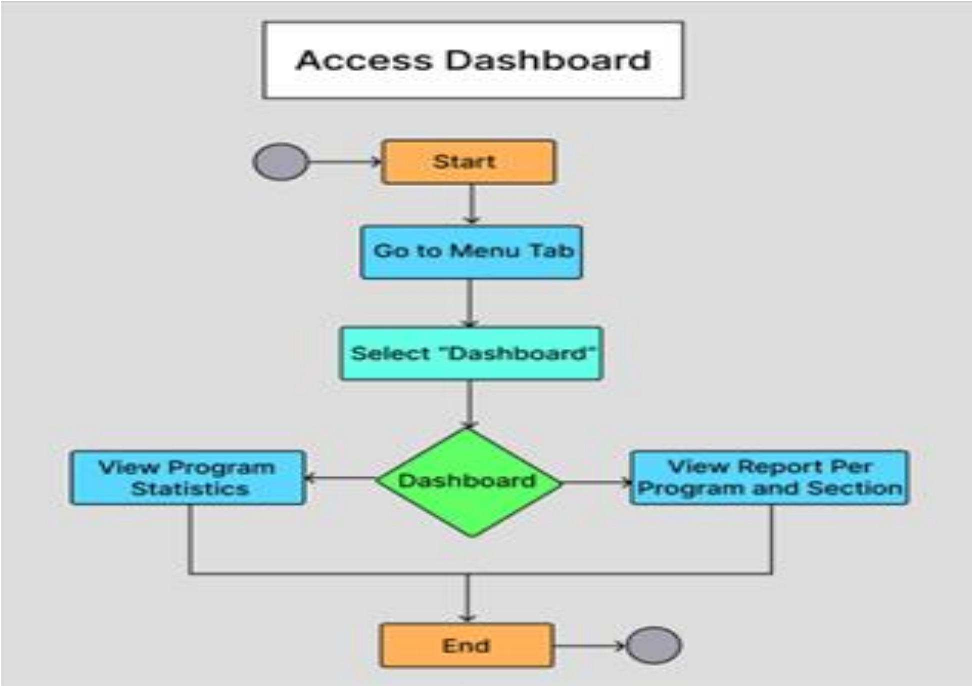

Figure 6. Coordinators Dashboard Activity Diagram

The activity diagram describes the process of the “Dashboard” feature within the OJT (On-the-Job Training) application for role of Coordinator and represents a steps sequence to be followed by a Coordinator. The sequence starts with activation, which is recognized with a starting point and the coordinator moves to a menu tab within the application interface. Next, the Coordinator clicks "Dashboard” and is presented with a decision point where they may choose one of two paths based on the purpose of their role: they may view program statistics, which takes them to statistical data about the specific OJT programs (e.g., hotspot, above). Each branch terminates with an "End" function doing completed action of the selected job and then, a termination point allowing the coordinator to effectively administer the OJT program data as needed.

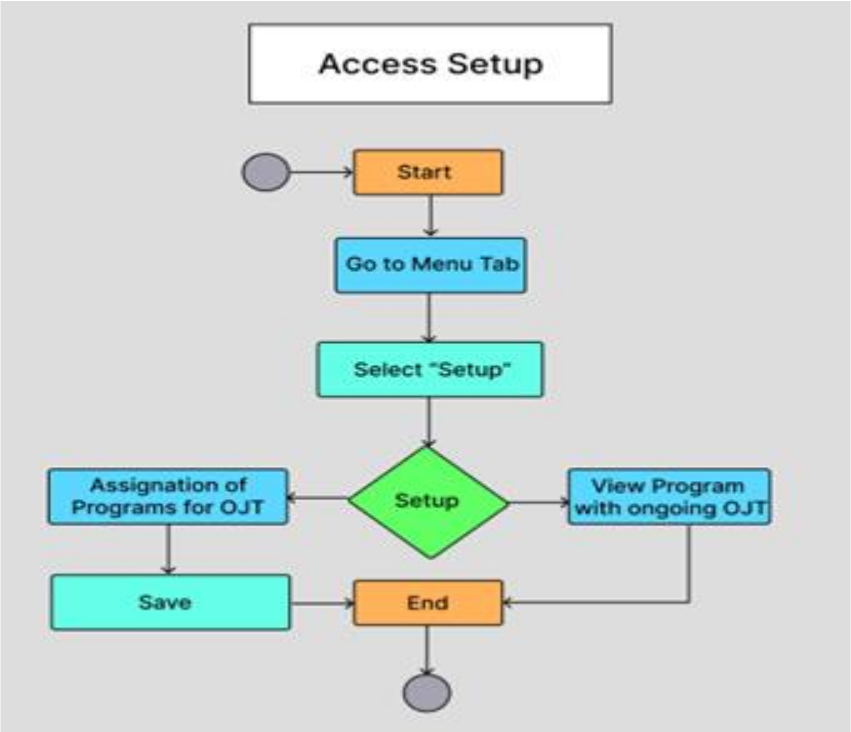

Figure 7. Coordinators Setup Activity Diagram

This "Setup" activity program illustrates the sequential steps involved the start process involves starting a task, and navigating to a menu tab, where a user can select 'Setup' to setup programs. From there the user can either assign programs for OJT and save and exit, or view programs that are already in progress with ongoing OJT and then exit. This flexible flow provides the coordinator with the option to create new assign or review existing ones before concluding after the selected assigned is finished, offering a straightforward and effective approach to OJT management.

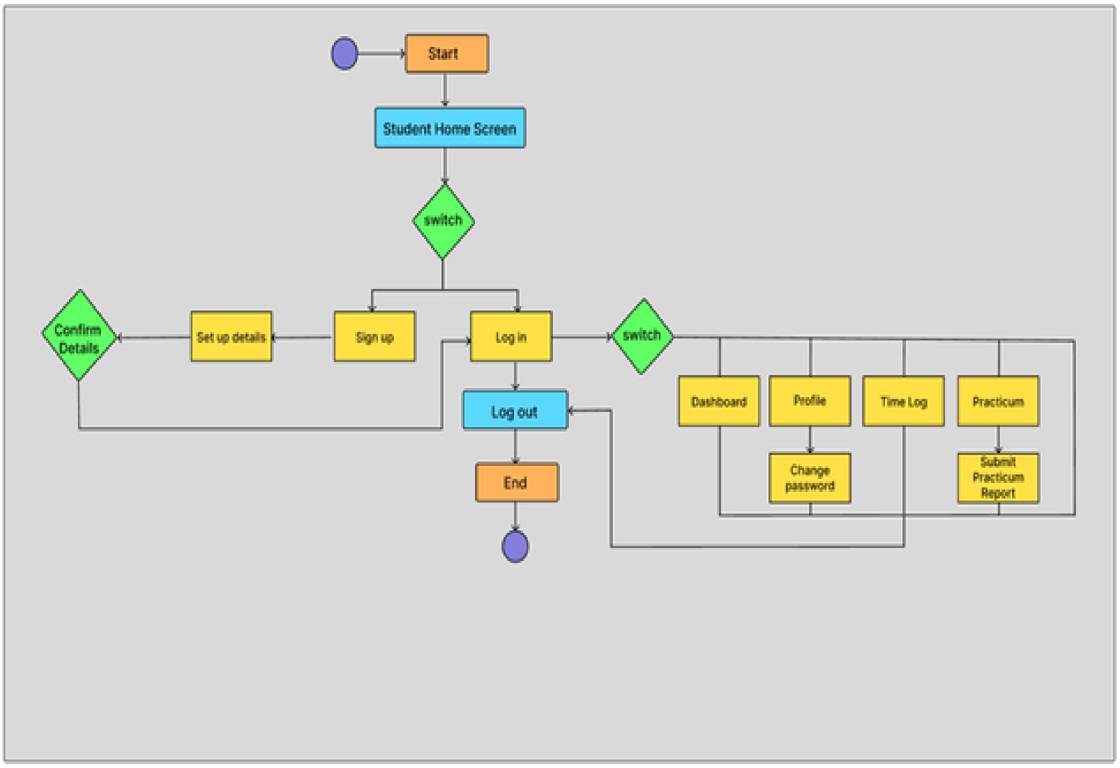

Figure 8. Students' Mobile Application System Use Case Diagram

The activity diagram illustrates the sequential steps starting with the process beginning and the student navigating to menu tab. From there, the student can choose the dashboard where the student can view hours overview which include the total hours rendered and remaining hours to render, the completion date, the progress of the practicum report of the student, also the student can view the calendar and can track days of the student's absences. The process wraps up once the student has completed these steps and ends, providing a simple and logical sequence for navigating to the dashboard.

3.2.3. GUI Design

Figure 9. OJT Coordinator Administrator View

For the Coordination View of OJT Application The process begins with a setup interface where users select a semester and choose a program from available degree and diploma courses. Following setup, the system allows for the management of practicum reports, where users can view and organize student reports. It includes a dashboard to track internship status, providing a visual representation of completion progress. Additionally, the system features a student performance section to monitor and update individual progress, and a login page to ensure secure access for authorized users. The overall workflow is supported by a centralized dashboard that offers an overview of intern totals and ongoing programs, facilitating efficient oversight and management.

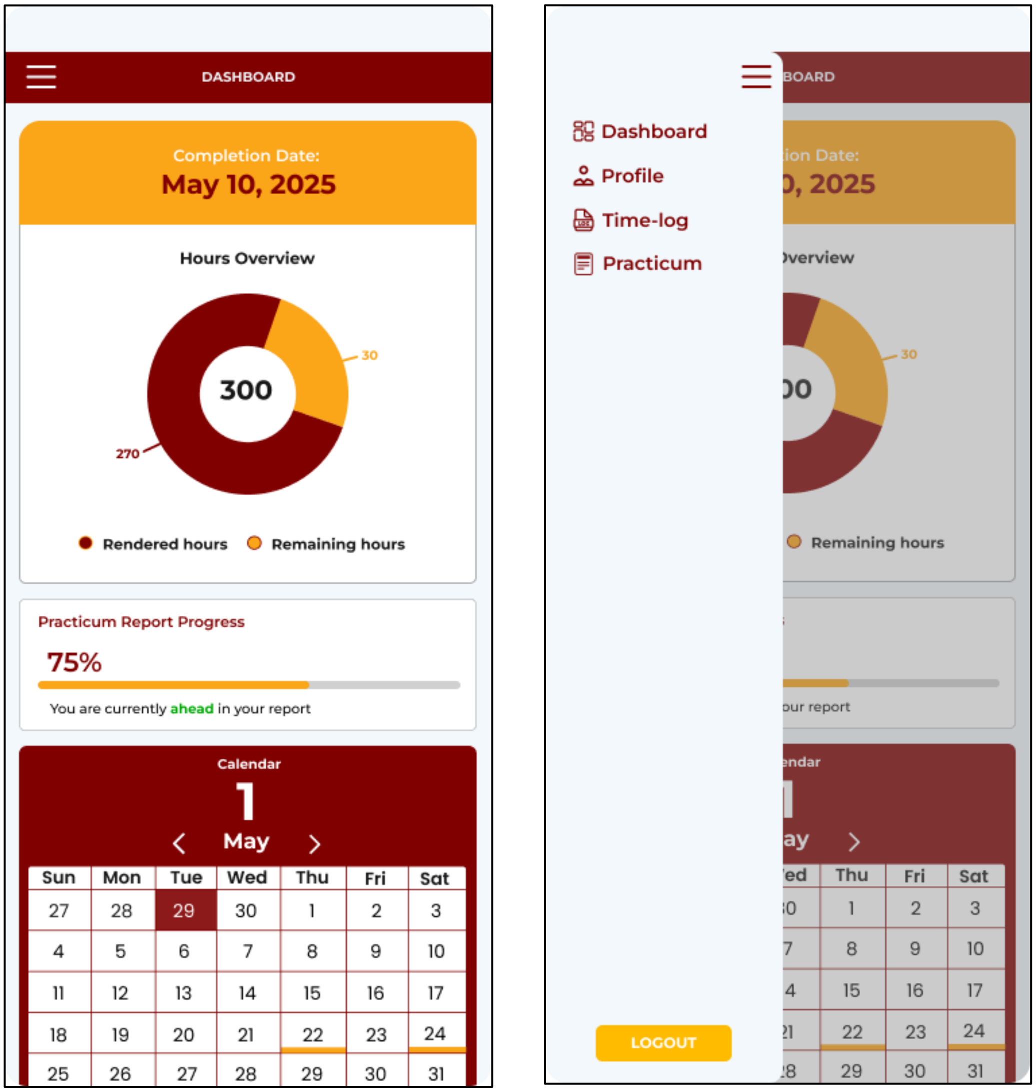

Figure 10. Students' Mobile Application View

The Student View on the OJT Application System cannot view the coordinator's web view except the Dashboard that shows student's completion date. But the Student can see other sections of the application that is compile using the hamburger menu. The Student can see the Student's Profile it can edit the profile picture, update the other information, can see the Student's Time-log, it can record the time-in and time-out of the student with real time capturing of an image, the student can also view the Practicum where the student can input all the necessary data and attach files needed, it can also edit the data of the practicum and save the changes and can also log out from the application.

3.2.4. Database Schema

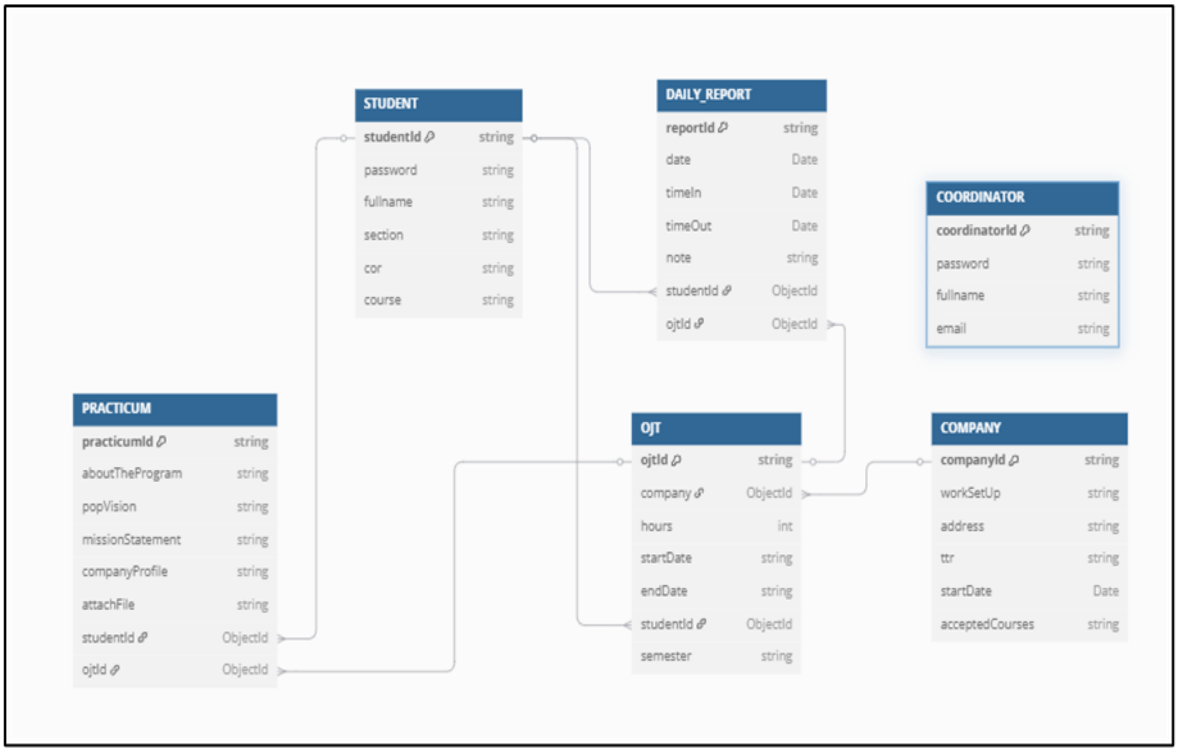

Figure 11. On-The-Job System Database Schema

3.2.5. Data Dictionary

Column Name	Data Type	Constraints	Description
companyId	string	PK	Unique company identifier
workSetUp	string		Remote/Hybrid/On-site
address	string		Company address
ttr	string		Training representative
startDate	Date		Training start date
acceptedCourses	string		Comma-separated courses accepted

Figure 12. Company Table Data Dictionary

Column Name	Data Type	Constraints	Description
coordinatorId	string	PK	Unique coordinator identifier
password	string		Encrypted password
fullname	string		Full name of coordinator
email	string	UNIQUE	Email address

Figure 13. OJT Coordinator Table Data Dictionary

Field Name	Data Type	Constraints	Description
studentId	string	PK	Unique student identifier
password	string		Student login password
fullname	string		Full name of student
section	string		Section name/label
cor	string		Certificate of Registration
course	string		Course title

Figure 14. Student Table Data Dictionary

Column Name	Data Type	Constraints	Description
reportId	string	PK	Unique report ID
date	Date		Report date
timeIn	Date		Time-in stamp
timeOut	Date		Time-out stamp
note	string		Work log note
studentId	ObjectId	FK	Reference to studentId from student table
ojtId	ObjectId	FK	Reference to ojtId from table OJT

Figure 15. Daily Report Table Data Dictionary

Column Name	Data Type	Constraints	Description
practicumId	string	PK	Unique practicum identifier
aboutTheProgram	string		Overview of the practicum
popVision	string		Vision statement
missionStatement	string		Mission statement
companyProfile	string		Profile description
attachFile	string		File attachment path
studentId	ObjectId	FK	Reference to studentId from table Student
ojtId	ObjectId	FK	Reference to ojtId from table OJT

Figure 16. Practicum Report Table Data Dictionary

Column Name	Data Type	Constraints	Description
ojtId	string	PK	Unique OJT record ID
company	ObjectId	FK	Reference to companyId from Company table
hours	int		Number of hours required
startDate	string		OJT start date
endDate	string		OJT end date
studentId	ObjectId	FK	Reference to studentId from Student table
semester	string		Semester label

Figure 17. OJT Table Data Dictionary

3.3. DEVELOPMENT METHODOLOGY

3.3.1. Process Model

The Incremental Model is a software development approach where the system is divided into multiple standalone modules, each module goes through requirements, design, implementation and testing phase. Every new version adds additional features to the one before it, and this process continues until the complete system is accomplished.

In the proposed On-The-Job Training (OJT) System Featuring a Centralized Company Hub with Student Completion Date Prediction and Practicum Report Generation for the Polytechnic University of the Philippines-Lopez Campus, the researchers effectively implemented the "Incremental Methodology" to address the limitations of traditional development methods. This approach enabled iterative progress, allowing the researchers to refine and improve the system while reducing the potential errors, producing a strong, flexible, and user-focused solution aligned with modern software engineering practices.

The process of the incremental software methodology used by the researchers to develop the OJT system involving several important steps. Firstly, the researchers identified the specific objectives and requirements, defining the system's purpose such as managing internship program, tracking student progress, and generating practicum reports and the problems it aims to solve. Next, the researchers conducted a comprehensive analysis of existing OJT management system to identify gaps and challenges. The researchers designed a basic version of the software or prototype, focusing on features that are more needed, which served as the foundation of further development. This initial version's code, design decisions, and functionalities were carefully documented by the researchers to ensure clarity and maintainability.

In incremental iteration, the researchers added new features such as time-in/out tracking, practicum report generation, and internship setup. Following each iteration, the researchers carefully documented any changes, pointed out any possible problems or bugs, and recorded the implemented solutions.

The researchers maintained clear and accurate documentation on the interfaces and system architecture. To make sure that the documentation was accurate and comprehensive, they reviewed and verified it at the end of each increment. By following the incremental methodology and thorough documentation, the researchers aimed to develop a reliable, interactive, and user-friendly OJT system designed to the needs of the Polytechnic University of the Philippines- Lopez Campus.

3.3.2. Development Tools

On-The-Job Training System is a digital system typed with Typescript for both front-end and back-end, ensuring type safety, code reliability, and maintainability. It uses Mongo DB as its primary database, efficiently managing user accounts, internship details, and practicum report. Visual Studio Code is used for development, writing, debugging, and managing codebases. It follows MERN Stack (Express.js, React.js, Node.js, and Mongo DB) and uses Typescript to ensure cohesive and type safety. Node.js and Express.js, along with TypeScript, handle back-end and API logic, while the front-end uses React Router V7 for smooth navigation and Tailwind CSS V4 for sleek, responsive designs. Adopting the Progressive Web App (PWA) design pattern, OJT web-app system delivers a native-like experience on mobile and desktop with enhanced performance. RESTful APIs enable smooth front-end and back-end integration, supporting features like image uploads and user authentication. OpenAI integration enhances the platform with AI-driven capabilities, such as automated weekly report summarization. The back-end and APIs are hosted on Heroku, while Firebase Hosting ensures fast, scalable front-end delivery. Cloudflare provides strong security, protecting the platform from threats. The domain is registered and managed via GoDaddy.

3.4. TEST METHODOLOGY/PROCEDURES

To ensure the reliability and functionality of the developed On-The-Job Training (OJT) System, a series of testing methodologies were employed. The development team used black box testing to evaluate the system's core features, such as account registration, company search, time logging, practicum report generation, and dashboard monitoring. Each function was tested without knowledge of the internal code, focusing on input-output validation.

User Acceptance Testing (UAT) was also conducted among selected students and the OJT Coordinator from PUP Lopez Campus to assess the system's usability and performance from an end-user perspective. Participants were provided with task scenarios and were asked to perform various operations using both the web platform and mobile application. Feedback was collected through surveys and direct observation to identify any bugs, usability issues, or improvement opportunities.

The testing process followed a structured procedure: identifying test cases based on system requirements, executing tests, recording results, and fixing any issues identified. The system was considered functional when all critical features operated without error, and the test participants were able to complete their tasks successfully.

3.5. SYSTEM REQUIREMENTS

The following system requirements must be in place to achieve a smooth, optimal experience at On-The-Job Training System. These will serve to ensure suitability, responsiveness, and accessibility by various devices.

Minimum System Requirements

Users are extremely encouraged to be at recommended specs for the full experience.

Category	Minimum Requirements
Device	Desktop, Laptop, Tablet, or Smartphone
Processor	Dual-core 1.5 GHz or equivalent
RAM	2 GB (Mobile) / 64 GB (Desktop) of available storage
Operating System	Windows 7 or later, macOS 10.12 or later, Android 7.0 or later, iOS 11 or later
Web Browser	Google Chrome, Mozilla Firefox, Safari, Microsoft Edge (latest versions)
Internet Connection	4G/5G or Wi-Fi with at least 5 Mbps speed

Table 4. Minimum System Requirements

Recommended System Requirements

Recommended specs allow for fast page loads and ease of navigation in On-The-Job Training System.

Category	**Recommended Requirements**
Device	Desktop, Laptop, Tablet or Smartphone
Processor	Quad-core 2.5 GHz or higher
RAM	4 GB (Mobile) / 8 GB (Desktop) or higher
Storage	32 GB (Mobile) / 128 GB SSD (Desktop) or higher
Operating System	Windows 10 or later, macOS 10.15 or later, Android 10 or later, iOS 14 or later
Web Browser	Latest versions of Google Chrome, Mozilla Firefox, Safari and Microsoft Edge
Internet Connection	High-speed internet (Fiber, 4G/5G, Wi-Fi with at least 20 Mbps speed)

Table 5. Recomended System Requirements

By meeting the requirements listed above, viewers will be able to view the On-The-Job Training System through an immersive digital experience, ensuring high-quality image rendering and smooth real-time navigation.

3.6. QUALITY PLAN

To ensure the developed *ON-THE-JOB TRAINING SYSTEM FEATURING CENTRALIZED COMPANY HUB WITH STUDENT COMPLETION DATE PREDICTION AND PRACTICUM REPORT GENERATION FOR THE POLYTECHNIC UNIVERSITY OF THE PHILIPPINES – LOPEZ CAMPUS* meets high standards of performance, a Quality Plan will be implemented based on the **FURPS** model: Functionality, Usability, Reliability, Performance, and Supportability. This model serves as a comprehensive framework for evaluating and validating the system's quality and effectiveness.

FURPS Evaluation Criteria

Functionality:

The system will be evaluated on its ability to:

- Display a list of partnered companies with MOA.
- Allow students to register, verify their Certificate of Registration (COR), and time-log their rendered hours.
- Enable automatic generation of completion dates and practicum reports.

Usability:

The design of the system will be checked for:

- Intuitive user interface and smooth navigation.
- Accessibility of key features like time-logging, report generation, and company search.

- Clear instructions for first-time users and non-technical users.

Reliability:

The reliability of the system will be assessed by testing:

- System performance across different devices and browsers (e.g., Android/iOS, Chrome, Safari).
- Stability under varying internet conditions (Wi-Fi and mobile data).
- Capability to recover from crashes or unexpected failures.

Performance:

Performance will be monitored through:

- Speed and responsiveness of rendering logs, company lists, and reports.
- Capability to handle multiple users logging time or submitting reports simultaneously.
- Load time and system behavior under high usage conditions.

Supportability:

Support and feedback mechanisms will be evaluated by checking:

- Availability of help sections or FAQs within the app.
- Responsiveness of the development/support team to bugs and inquiries.
- Ease of reporting issues and submitting feature requests.

Survey Tool for Quality Assessment

A structured survey form will be distributed to end-users (students, OJT coordinators) to collect feedback regarding their experience with the system. This survey includes demographic information, system usage patterns, and rating questions under the FURPS categories using a Likert Scale (1 = Strongly Disagree to 5 = Strongly Agree). Additional open-ended questions will gather qualitative insights and suggestions for future improvements.

Data Privacy and Ethics

All data gathered will be handled in accordance with ethical guidelines and data privacy laws. Participation in the survey is voluntary, and personal information is optional. All responses will be anonymized and used solely for improving the quality of the system.

Continuous Improvement

The collected feedback and testing results will be analyzed to identify areas for improvement. These insights will guide future updates, enhancements, and bug fixes, ensuring that the system remains aligned with the evolving needs of students and administrators.

3.7. EVALUATION PLAN

The evaluation of the On-The-Job Training System will be conducted to ensure that the system meets its intended objectives and provides functional, user-friendly, and efficient experience for students and OJT coordinators. The evaluation will be guided by the FURPS model, which includes Functionality, Usability, Reliability, Performance, and Supportability. A structured User Acceptance Testing (UAT) will be implemented involving actual users, such as students and OJT Coordinator, to gather relevant feedback. Surveys, interviews, and observation techniques will be used to assess system performance and user satisfaction. The feedback will be analyzed and used for system refinement and final validation. The success of the system will be determined by how well it addresses the initial problems identified, such as difficulty in finding Host Training Establishments, tracking rendered hours, and generating practicum reports.

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National

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APPENDICES

A. Problem Analysis Data Gathering, Questions and Responses

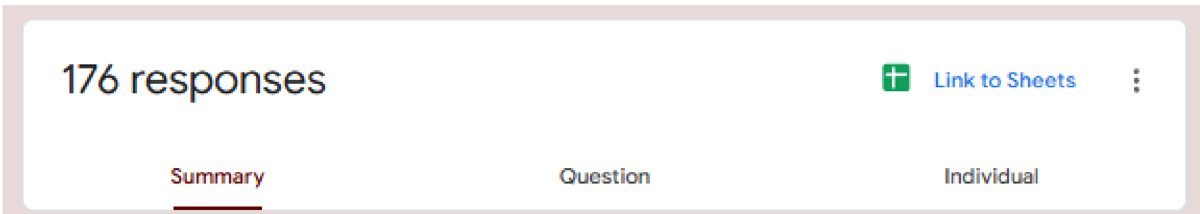

Figure 18. Total number of responses

Figure 19. Program/Year/Section of the respondents

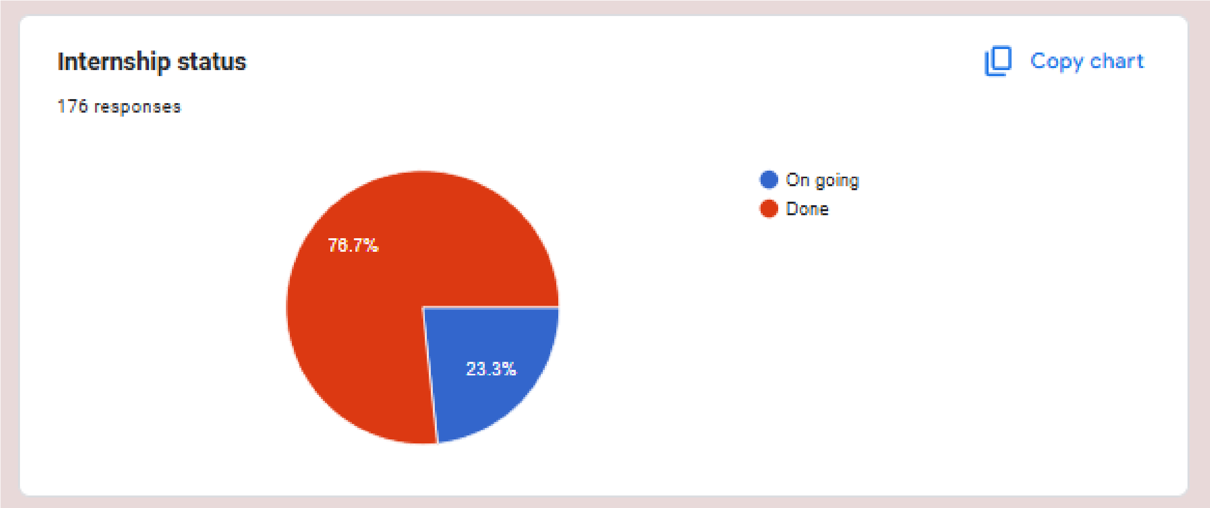

Figure 20. Internship status of respondents

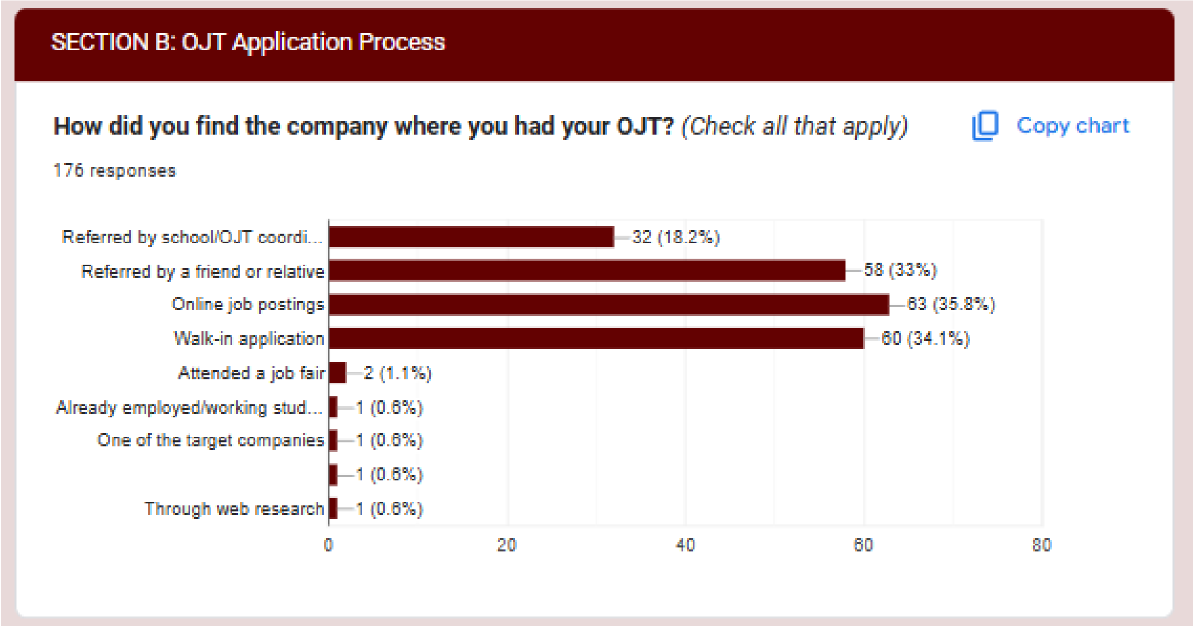

Figure 21. Responses to "How did you find the company where you had your OJT?"

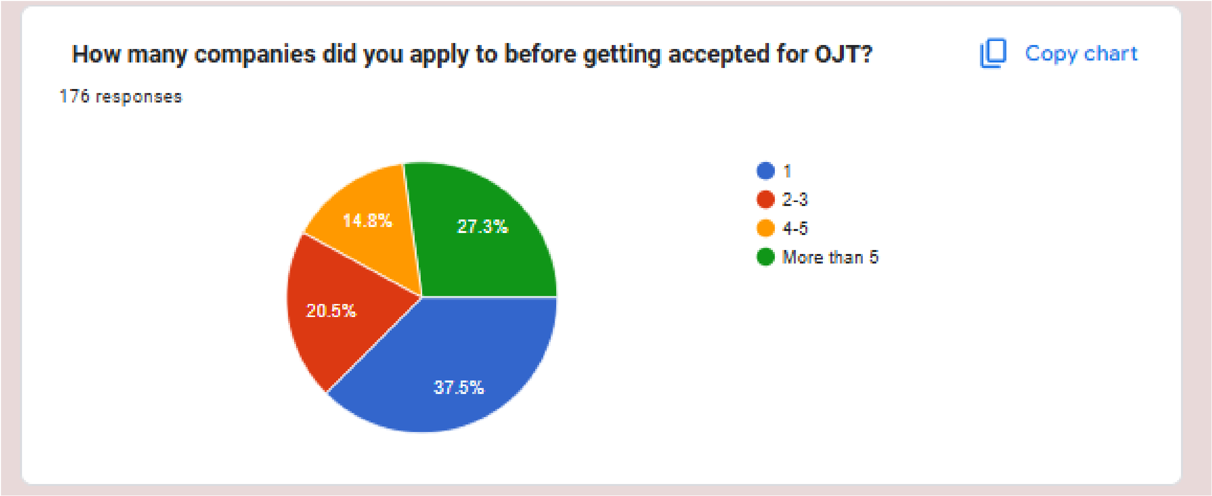

Figure 22. Responses to " How many companies did you apply to before getting accepted for OJT?"

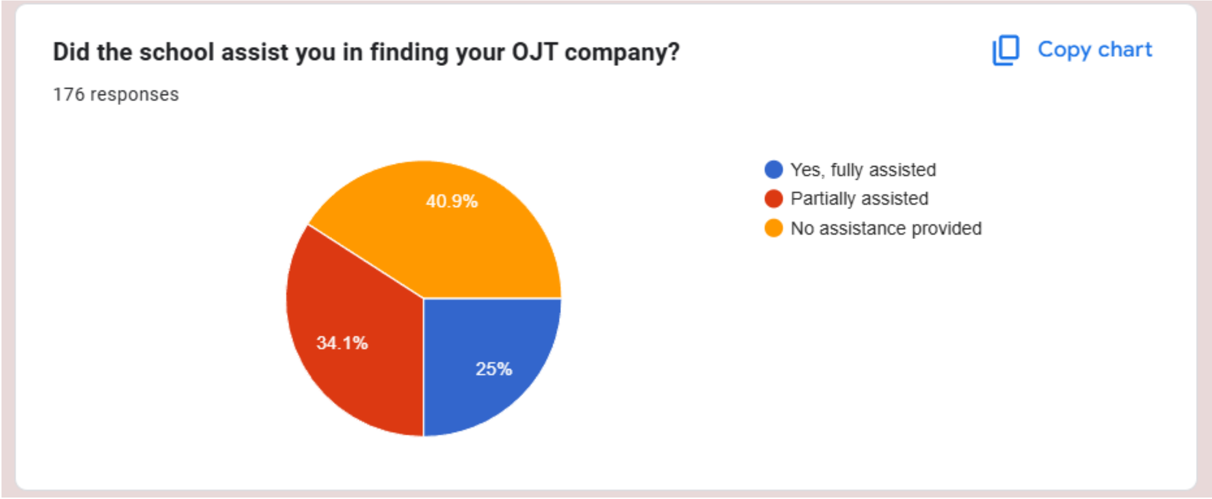

Figure 23. Responses to " Did the school assist you in finding your OJT company?"

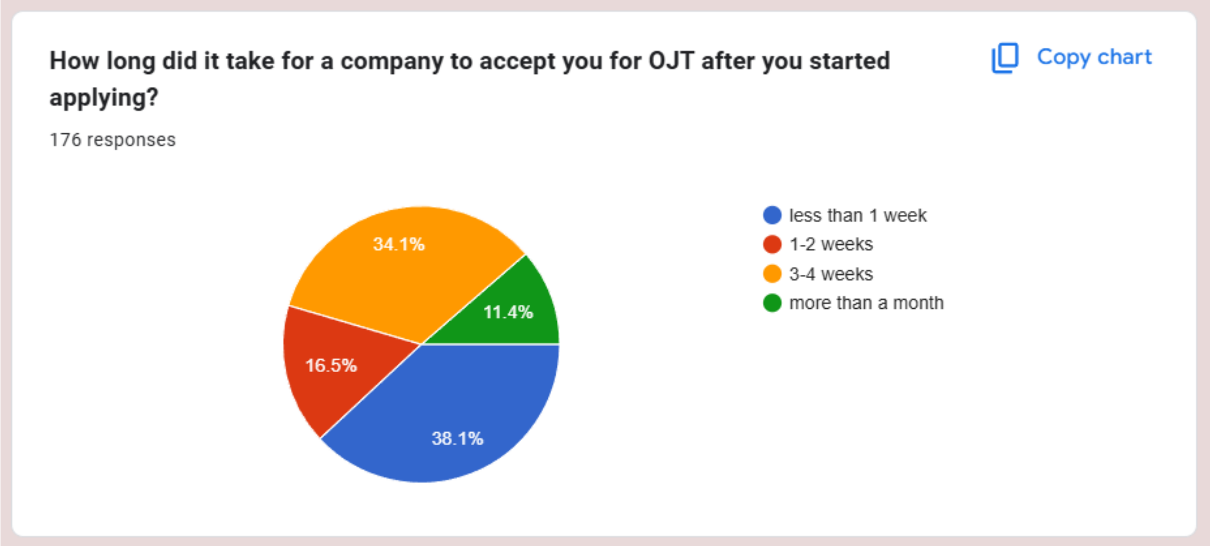

Figure 24. Responses to " How long did it take for a company to accept you for OJT after you started applying?"

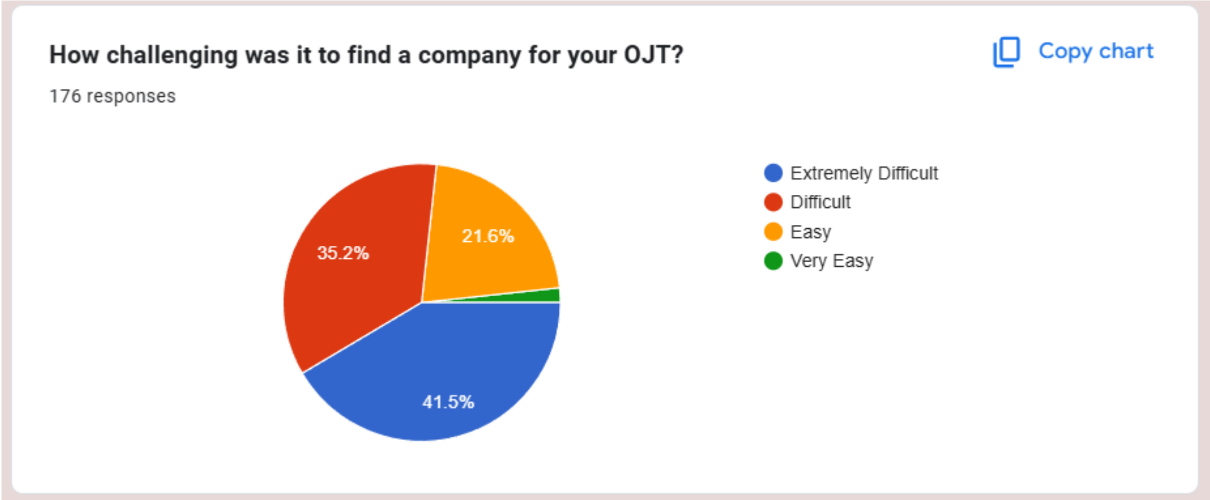

Figure 25. Responses to " How challenging was it to find a company for your OJT?"

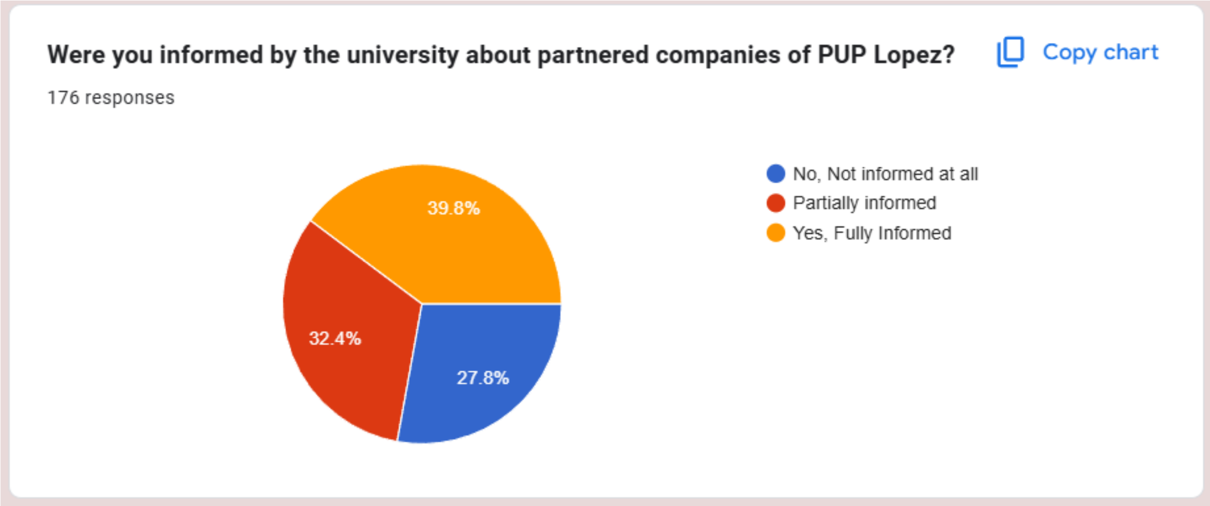

Figure 26. Responses to " Were you informed by the university about partnered companies of PUP Lopez?"

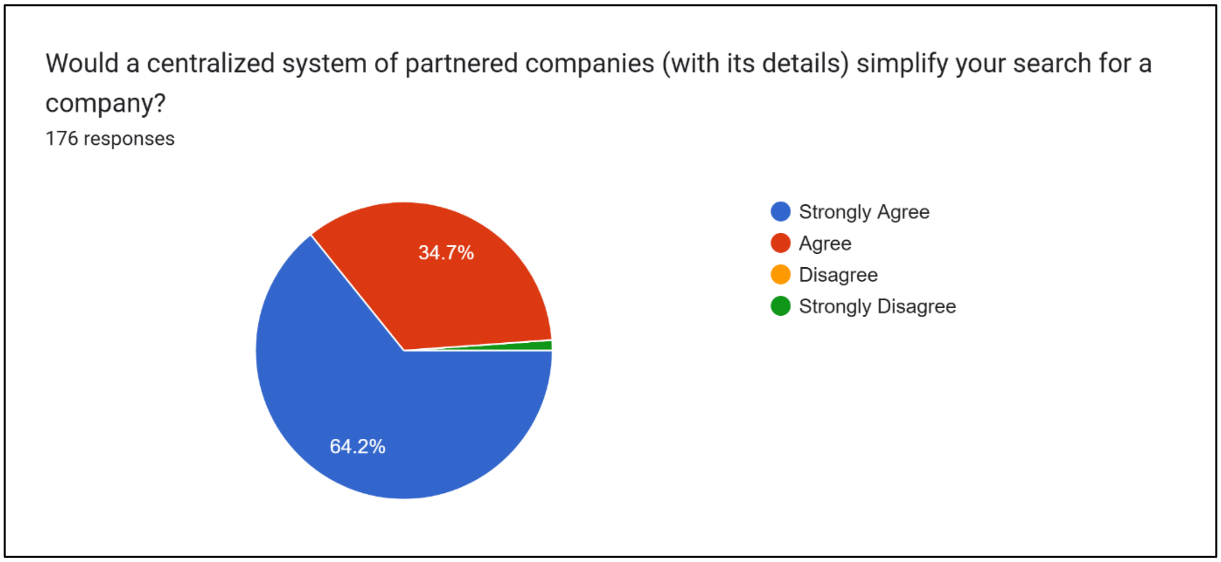

Figure 27. Responses to " Would a centralized system of partnered companies (with its details) simplify your search for a company?"

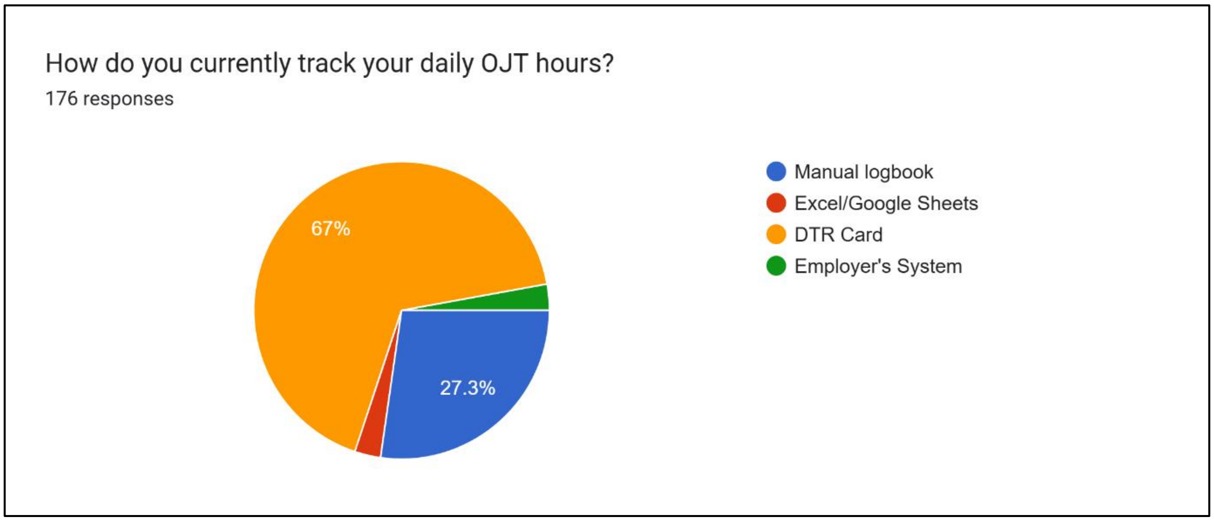

Figure 28. Responses to " How do you currently track your daily OJT hours?"

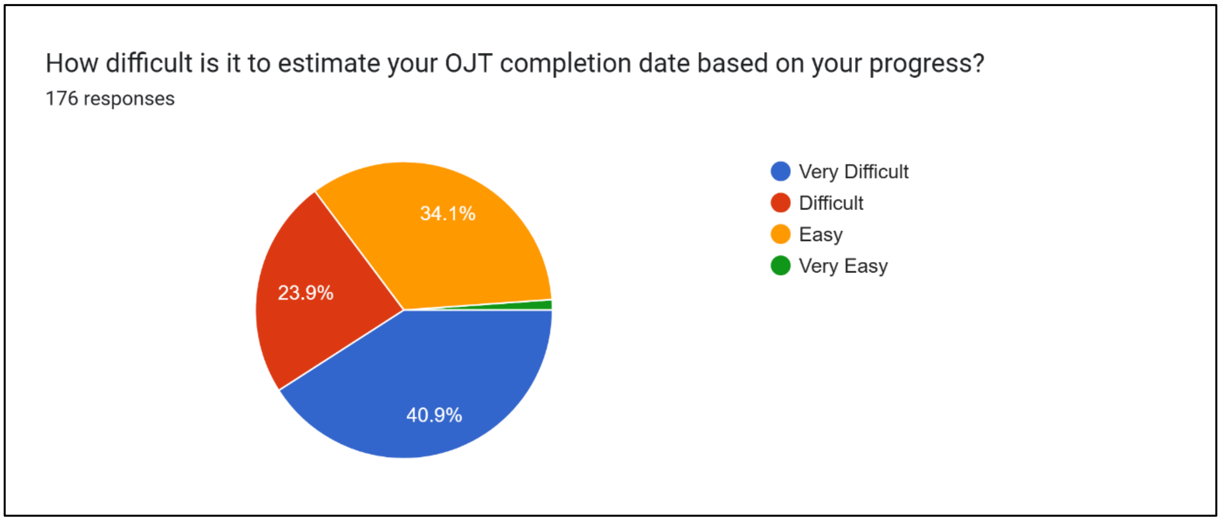

Figure 29. Responses to " How difficult is it to estimate your OJT completion date based on your progress?"

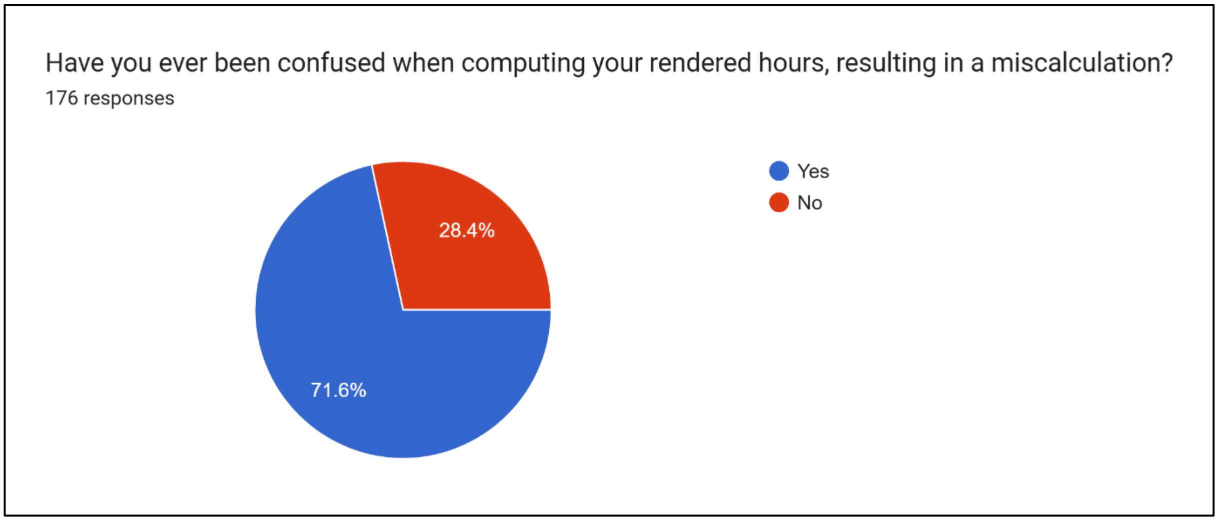

Figure 30. Responses to " Have you ever been confused when computing your rendered hours, resulting in a miscalculation?"

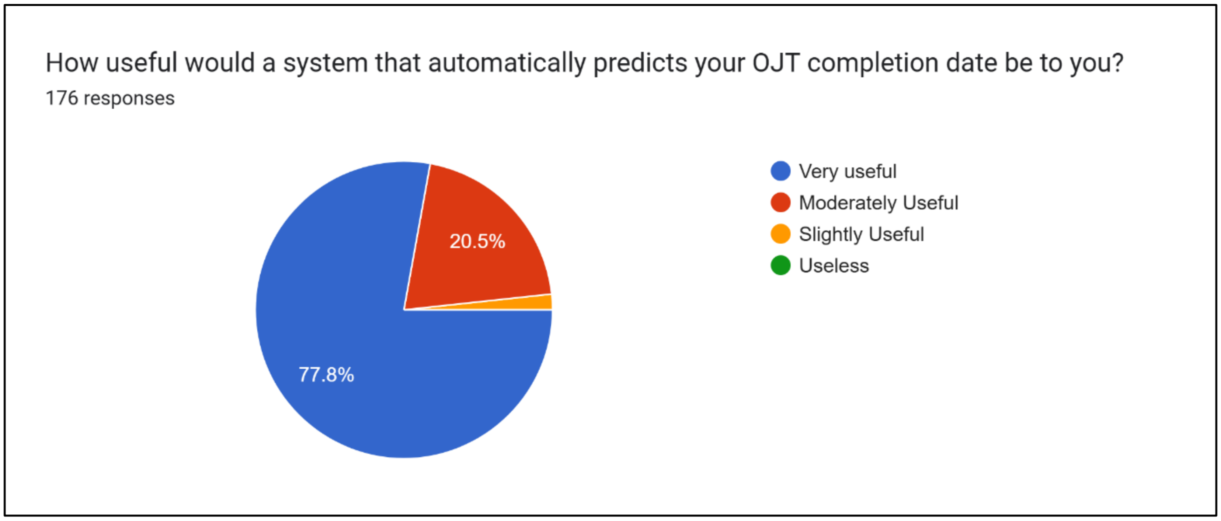

Figure 31. Responses to " How useful would a system that automatically predicts your OJT completion date be to you?"

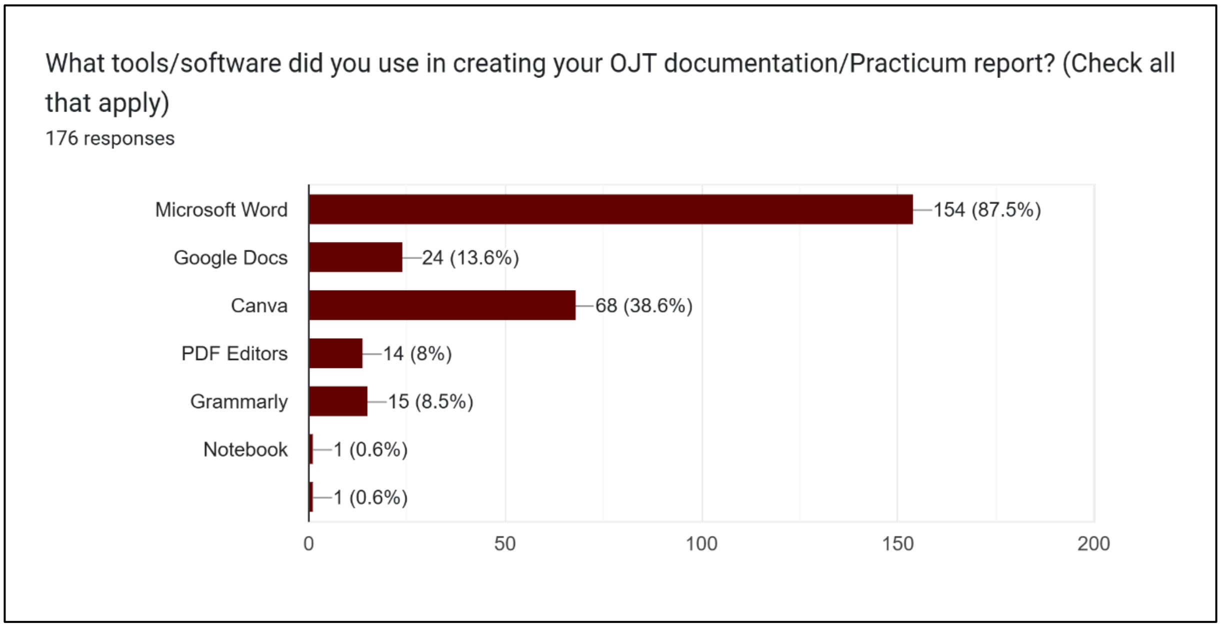

Figure 32. Responses to " What tools/software did you use in creating your OJT documentation/Practicum report? (Check all that apply)?"

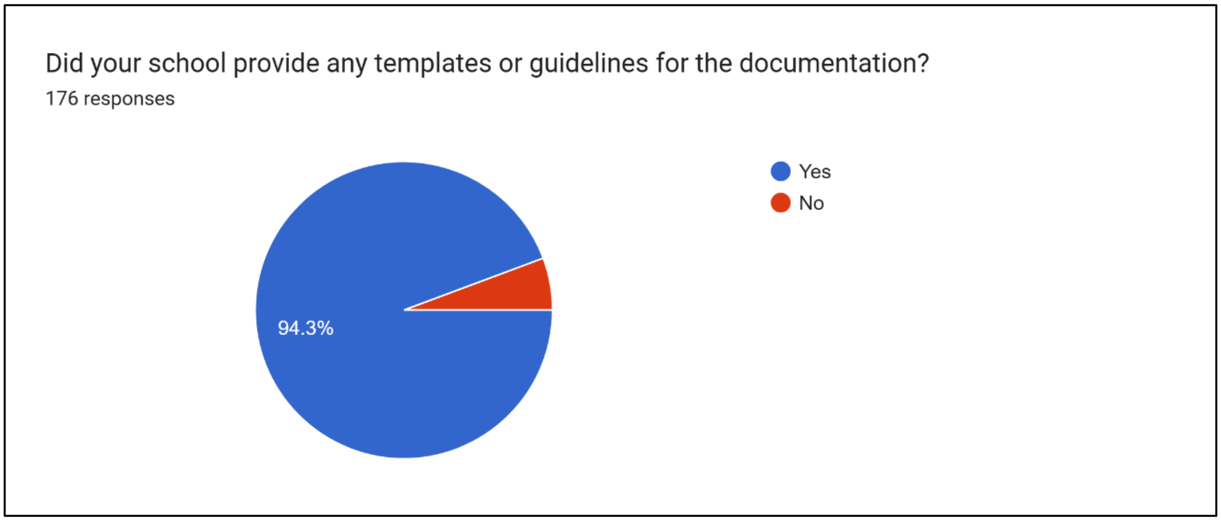

Figure 33. Responses to " Did your school provide any templates or guidelines for the documentation?"

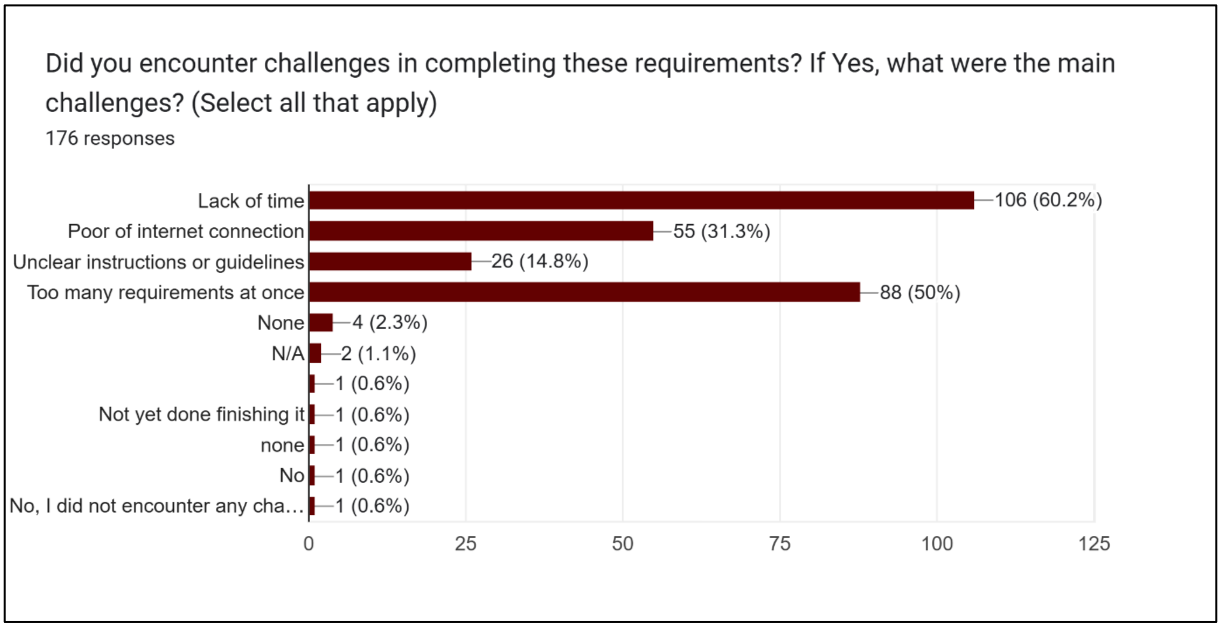

Figure 34. Responses to " Did you encounter challenges in completing these requirements? If Yes, what were the main challenges? (Select all that apply)?"

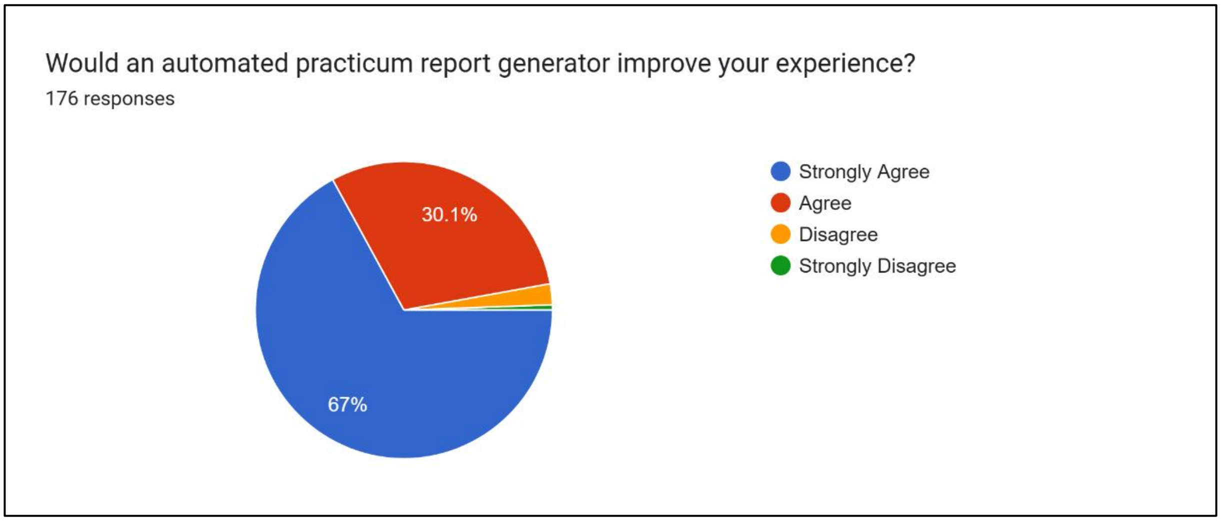

Figure 35. Responses to " Would an automated practicum report generator improve your experience?"

B. Curriculum Vitae

AJEMHER M. BALAHULA

IT STUDENT

ABOUT ME

Aspiring and motivated individual seeking an internship position as a Programmer, aiming to leverage my foundational knowledge in programming, problem-solving skills, and eagerness to learn in a dynamic and innovative environment. Dedicated to contributing to team projects, developing efficient code, and growing professionally within the software development field.

09105798502
ajemndnblhl@gmail.com
Bray, Castillo Pitogo Quezon

EDUCATION

Pitogo Community High School
2016 - 2022

Polytechnic University of the Philippines Lopez Campus
2022 - Present

SOFT SKILLS

- Leadership
- Teamwork
- Time Management
- Critical Thinking
- Written and Verbal Communication

TECH STACK AND PROGRAMMING LANGUAGES

MERN Stack	Java
Visual Basic	C++
Python	C#

INTERNSHIP EXPERIENCE

Fullstack Developer
Trifecta Solutions Inc.
July 29 2024 - Sep 06 2024

ACADEMIC ACHIEVEMENTS

President Lister
1st year, 1st semester

Dean Lister
1st year, 2nd semester

President Lister
2nd year, 1st semester

Dean Lister
2nd year, 2nd semester

LEADERSHIP AFFILIATION

PCHS - SSG P.I.O
Junior High, 2019 - 2020

IBITS ORG VICE PRESIDENT
2nd year, 1st semester, 2023 - 2024

PCHS - BDK VICE PRESIDENT
Senior High, 2022 - 2023

IBITS ORG PRESIDENT
2nd year, 2nd semester, 2023 - 2024

CLASS PRESIDENT
1st year, 2023 - 2024

CERTIFICATES AND SEMINARS

"Data analytics and how to become a data analyst and creating a video presentation, and Developing a Mobile Cross-Platform Application"
January 31, 2023

Research Colloquium "Contextualizing Academic Researchers in the Framework of SGDs"
June 04, 2024

"Quezon Cybersecurity Conference and Chapter activation"
June 29, 2024

"Certified member of Philippine Institute of Cyber Security Professionals"
June 15, 2024

"16th EdukCircle International Convention On Engineering and Computer Technology"
May 18, 2024

MARTINNE L. TAÑADA

IT STUDENT

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Gumaca, Quezon Philippines
07martimmetanada@gmail.com

ABOUT ME

As a third-year IT student at PUP Lopez, Quezon seeking for internship opportunities I am passionate in UI/UX designing with a keen eye for detail and a strong drive to create intuitive, user-friendly experiences. I thrive on learning new things, especially when they spark my curiosity, and I'm always eager to expand my skill set. As a good listener, I value feedback and collaboration, ensuring that every project is thoughtfully designed and well-executed. With a deep appreciation for aesthetics and functionality, I strive to craft seamless digital experiences that leave a lasting impact.

EDUCATION

POLYTECHNIC UNIVERSITY OF THE PHILIPPINES LOPEZ QUEZON
Diploma in Information Technology
2022 - Present

GUMACA NATIONAL HIGHSCHOOL
2016 - 2021

GUMACA WEST ELEMENTARY SCHOOL
20011 - 2016

SKILL

- Visual Imagination
- Communication
- Typography
- Attention to details

EXPERTISE

- UI/UX Design
- Graphic Design
- Visual Basic
- C++
- C#

LANGUAGE

- Tagalog
- English

WORK EXPERIENCE

2024-2024 **STRONGHOLD INSURANCE COMPANY INC**

UI/UX Designer

In 2024, during my On the Job Training (OJT) I had the incredible opportunity and become an intern student at Stronghold Insurance Company in the IT Department, there I worked as a UI/UX and graphic designer for their system. During my time in Stronghold, I was responsible for designing intuitive user interfaces and creating visually engaging graphics that enhanced the overall user experience. This experience allowed me to apply my creativity and problem-solving skills in a real-world setting while collaborating with a team of developers and IT professionals. I gained valuable hands-on experience in refining designs based on user feedback, ensuring that the systems were both functional and aesthetically pleasing. This internship not only strengthened my technical skills but also reinforced my passion for UI/UX design, motivating me to continue learning and growing in the field.

ACADEMIC ACHIEVEMENTS

- With Honors
  Senior Highschool
- With Honors
  Junior Highschool

CERTIFICATES

- Contextualizing Academic Research in the Framework of SDG's
  June 4, 2024
- EdukCircle International Conversation On Engineering and Computer Technology
  May 18, 2024
- PicsPro Quezon Chapter CyberSecurity Conference
  July 4, 2024

Arianne E. Gupa

Calauag, Quezon, Philippines
Mobile: +63 977 057 6054
Email: gupaarianne@gmail.com

About

An aspiring and dedicated IT student seeking an opportunity to this internship to gain more knowledge that could add and enhance my skills. Shove herself to quickly adopt to the environment that is new to her. Ready to face challenges, to contribute and learn from future team project to improve my skills and knowledge in IT field.

Educational Background

Secondary Education
Sto. Domingo National High School

Tertiary Education
Polytechnic University of the Philippines
Lopez Campus

Skills

C++
Visual Basic
Python
Java

Soft Background about React, Node.js and Express.js, MongoDB, and Postman for testing API

Internship Experience

During my internship at Trifecta Solutions Inc. as a Junior Programmer, I was assigned as a Backend Developer and been part of the Frontend Developer team also. I gained practical experience in analyzing and debugging real-world situations through collaboration with my teammates. This experience helped me to improve my knowledge, which also gave me the courage to use what I had learned during my internship to my future workplace. The experience I gained prepared me for this upcoming internship and future project with my team.

Academic Achievements

With Honor
Grade 12

President Lister
1st Year 1st Semester
2nd Year 1st Semester

Dean Lister
1st Year 2nd Semester
2nd Year 2nd Semester

Leadership Affiliation

Class Vice President
(1st Year, 2022-2023)

Class President
(2nd Year, 2023-2024)

STEPHANIE LUCERO

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EDUCATION

Tertiary Education
Polytechnic University of the Philippines Lopez Campus, 2022 - Present

Secondary Education
Sisters of Mary of Bannuex, Inc., 2015

ACADEMIC ACHIEVEMENTS

With Honor
Grade 10

With Honor
Grade 11

With Honor
Grade 12

President Lister
1st Year , 1st Semester

President Lister
1st Year, 2nd Semester

President Lister
1st Year, 1st Semester

Dean Lister
1st Year, 2nd Semester

ABOUT ME

As a dedicated IT student seeking an internship opportunity to gain practical experience, enhance my skills, and deepen my knowledge in real-world IT environments. I am driven by a strong desire to learn and thrilled about the potential to progress in the exciting sector of information technology.

WORK EXPERIENCE

I completed my first On-The-Job Training at Trifecta Solutions, Inc in Quezon City from July 2024 to September 2024. I gained experience in frontend, UI design, and backend coding while developing a web application. I got hands-on experience with React, attended workshop, and helped with project planning and task assignments. Along the way, I faced challenges like debugging issues and time constraints. Despite the difficulties, I kept in moving forward while collaborating closely with my team. Even though there were some delays, I remained dedicated to enhancing my job and completing everything as efficiently as possible.

PROGRAMMING SKILLS

- C++
- Java
- C#
- Visual Basic
- MySQL
- Python

TECHNICAL SKILLS

- Figma
- Postman
- Microsoft Office
- Visual Studio Code
- Microsoft Visual Studio
- Python

IT STUDENT

Guinayangan, Quezon
jjpasorren99@gmail.com
+9634782260

EDUCATION

Secondary Education
2016 – 2022
Aloneros National High School

Tertiary Education
2022 – Present
Polytechnic University of the Philippines - Lopez Campus

TECHNICAL SKILLS

- Microsoft Office
- Visual Studio Code
- Figma
- GitHub
- Postman

Joseph John Paul A. Almazan

ABOUT ME

Motivated IT student in my third year at Polytechnic University of the Philippines, Lopez Quezon Branch. Seeking an OJT opportunity to apply my technical skills and gain real-world experience in software development, network administration, or cybersecurity. Passionate about learning new technologies and working in a fast-paced and dynamic environment. Excited to contribute my ideas and learn from experienced professionals to further my career in IT

INTERSHIP EXPERIENCE

I completed my first On-the-Job Training (OJT) at Trifecta Solutions Inc. in Quezon City from July 2024 to September 2024. During my time there, I served as a Junior Programmer 1/Junior Programmer 2, where I was responsible for UI designing, frontend developing, and backend developing. Throughout the internship, I successfully completed tasks within deadlines and developed skills in UI design, frontend, and backend development. The experience provided me with valuable insights into the industry and allowed me to apply my technical skills in a real-world setting. I am grateful for the opportunity to have worked with a team of experienced professionals and to have gained practical experience in software development.

PROGRAMMING SKILLS

Frontend Development

- **React**: Familiar with React and frontend development
- **TypeScript**: Familiar with TypeScript programming concepts

Backend Development

- **Node.js & Express:** Familiar with developing scalable RESTful APIs and backend services.
- **MongoDB:** Familiar with designing schemas

ACADEMIC ACHIEVEMENTS

With Honor

- Grade 12

Dean's Lister

- 1st year, 1st semester
- 1st year, 2nd semester
- 2nd year, 1st semester
- 2nd year, 2nd semester